

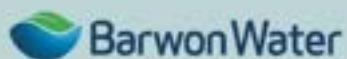
Annual Drinking Water Quality Report

2023/24





Further information



For more information on the 2023/24 Annual Drinking Water Quality Report, or any other concerns regarding water quality, including opportunities to work with us on improving water quality services, please contact our Product Quality Team on 1300 656 007 or email us at info@barwonwater.vic.gov.au.

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Message from the Managing Director



On behalf of Barwon Water, I am delighted to introduce our Annual Drinking Water Quality Report for 2023/24.

We operate on the lands of Wadawurrung and Eastern Maar. We proudly acknowledge them as the Traditional Owners of the land and water on which we rely, and pay respects to their Elders, past, present and emerging. We also acknowledge that Wadawurrung and Eastern Maar never ceded this land or the water that flows through it. We thank them for their enduring care of the land, water and the natural environment over tens of thousands of years, a stewardship that continues to this day.

Our Annual Drinking Water Quality report outlines our ongoing commitment to provide safe, secure, high-quality drinking water to our community.

Over this financial year, we have taken significant steps towards fulfilling our vision set out in Strategy 2030 and our 2023-2028 Price Submission. This vision outlines our actions, in light of the challenges presented by a changing climate and growing region.

During this five-year pricing period, we have committed to investing nearly \$95 million in improving drinking water quality, including through upgrades to water treatment plants, our asset renewals program, catchment management, and preventive asset management initiatives.

Central to our strategy is the understanding that a healthy environment is crucial to safeguarding drinking water quality. Our investment and partnerships in various initiatives not only improve the quality of water in our catchments, but also support the vitality of our waterways. By prioritising these initiatives, we prioritise the wellbeing of our customers, the community, and the natural environment.

Despite challenging environmental conditions in 2023/24, including an unprecedented blue-green algae bloom at Wurdee Boluc Reservoir, we maintained consistently high-quality drinking water.

While there was a slight increase in drinking water complaints in the financial year, we are proud that we have been able to maintain this number at near record lows. This is a testament to our continuous improvement focus and high levels of community trust.

Our commitment to the supply of safe, secure, sustainable water is outlined in our Drinking Water Quality Policy. We are also focused on being accountable, and this report provides a guide to our performance.

It is with pleasure that I present Barwon Water's 2023/24 Annual Drinking Water Quality Report.

A handwritten signature in blue ink, appearing to read 'Shaun Cumming'. The signature is fluid and stylized, with a long horizontal stroke extending to the right.

Shaun Cumming
Managing Director
Barwon Water

Key information

12 major reservoirs	8 surface water treatment plants	1 pre-treatment plant for ground water operations
22 secondary disinfection plants	26 water distribution basins	49 water distribution tanks
7,597 kilometers of water mains	8,100 square kilometers serviced	30 water quality localities
379,228 permanent population serviced	579,300 peak population serviced	181,844 properties serviced

1 Introduction

In accordance with the *Safe Drinking Water Act 2003*, this report provides an overview of Barwon Water's water supply system, the risk management processes in place to ensure the provision of safe drinking water and water quality performance for the 2023/24 financial year.

Barwon Water supplied approximately 37,229 million litres of drinking water during 2023/24 through a network of 12 reservoirs, 1 groundwater borefield, 8 water treatment plants, 22 disinfection sites and 7,597 kilometres of pipes.

To ensure customers are provided with safe drinking water, Barwon Water maintains a HACCP-certified drinking water quality management system, which covers the entire water supply system from the catchment to customers' connection. This system entails:

- Detailed water quality risk identification processes
- Audited control measures to manage the risks
- Verification systems, including an independent water quality monitoring program
- Water quality performance reporting to the public
- Continuous improvement projects to further increase the systems capabilities.

Throughout 2023/24, we delivered drinking water to all 30 water sampling localities in accordance with the required water quality standards in the *Safe Drinking Water Regulations 2015*.

2 Requirements for Drinking Water

2.1 Meeting customer expectations

Barwon Water is committed to managing its water supply effectively to provide safe, high quality drinking water to its customers and ensure public health. This commitment forms part of Barwon Water's Drinking Water Quality Policy and our Customer Charter.

To ensure Barwon Water is open to customer feedback, the following measures are in place:

- Complaints management system – this is used to record all complaints received regarding water quality.
- Customer perception surveys – a broad survey carried out to ascertain the general customer attitude towards the quality of service provided.
- Customer Consultative Committee – a committee formed of members of the community & Barwon Water customers.

- Customer feedback received during community panel workshops and community engagement events.

2.2 Legislative requirements

The *Safe Drinking Water Act 2003* (the Act) is the principal legislation applicable to the quality of drinking water supplied by Barwon Water. The Act:

- requires Barwon Water to prepare, implement, review and revise plans to manage risks in relation to drinking water and some types of non-potable water
- provides criteria for the auditing of those plans by approved auditors
- requires Barwon Water to ensure drinking water meets quality standards specified in associated regulations (*Safe Drinking Water Regulations 2015*)

- requires Barwon Water to disclose to the public information concerning the quality of drinking water.

The Health (Fluoridation) Act 1973 regulates the safe and effective addition of fluoride into drinking water supplies in Victoria. *The Health (Fluoridation) Act 1973* requires Barwon Water to include the Code of Practice for the Fluoridation of Water Supplies in their quality management system.

3 Defining drinking water

Drinking water supplied by Barwon Water must meet the obligations under the *Safe Drinking Water Act 2003* and *Safe Drinking Water Regulations 2015*.

The Australian Drinking Water Guidelines 2011 (ADWG) are used as a benchmark for safe drinking water in Australia. These guidelines are intended to provide a framework for effective management of drinking water supplies that, if implemented, will assure safety at point of use.

The ADWG include two types of guideline values:

- A health-related guideline value, which is the concentration or measure of a water quality characteristic that, based on present knowledge, does not result in any significant risk to health of the consumer over a lifetime of consumption.
- An aesthetic guideline value, which is the concentration or measure of a water quality characteristic that is associated with acceptability of water to the consumer, such as appearance, taste and odour.

2.3 Undertakings, Variations and Exemptions

Barwon Water had no undertakings pursuant to regulation 16(c) of the *Safe Drinking Water Regulations 2015*, during this reporting period.

No variation in aesthetic standards pursuant to regulation 16(i)(i) or exemptions from a water quality standard pursuant to regulation 16(i)(ii) of the *Safe Drinking Water Regulations, 2015* were sought during this reporting period.

The intended use of drinking water supplied by Barwon Water is:

- immediate consumption by the general public, with no further treatment or boiling by the consumer necessary for it to be reasonably considered safe and aesthetically acceptable.
- other domestic and commercial uses where the requirements for these other applications do not exceed government obligations.

The ADWG states that “The Guidelines are derived so as to take account of the needs of an individual through a normal lifetime, including changes in sensitivity that may occur between life stages” and that “Sensitive sub-populations (including those who are severely immuno-compromised) should seek further medical advice”.

The water quality data presented in this report is representative of the drinking water supplied in accordance with this definition and the above-mentioned requirements.

3.1 Regulated Water

Regulated water is “...water that is not intended for drinking water but which could reasonably be mistaken as being drinking water.” (as defined in the Department of Health, *Regulated Water – Drinking Water Regulation Guidance Note*).

Barwon Water does not have any regulated water supplies.

4 System Description

4.1 Overview

Barwon Water provided drinking water to a permanent population of approximately 379,228 people across an area of more than 8,100 square kilometres.

A map of the water supply system, including an overview of Barwon Water's service area and major infrastructure, is provided in Figure 1.

Source water is harvested from water supply catchments and is stored in reservoirs.

Additional water was also accessed from an interconnection between Geelong and Melbourne, the Melbourne to Geelong Pipeline.

Groundwater from the Anglesea borefield was not utilised during 2023/24.

A network of pipes, channels and pumps then conveys the water to treatment plants, where it is treated to produce drinking water. The water is then delivered to customers by a network of pipes, basins, tanks and pump stations.

Figure 1 Barwon Water service area and major infrastructure



4.2 Water quality localities

Barwon Water's reticulation system is divided into water quality localities for the purpose of monitoring and reporting. Allocation of water quality localities allows for greater consistency in defining quality zones. This provides more meaningful data analysis, a greater understanding of water quality issues directly related to treatment,

consistency in water quality monitoring / reporting and a more intelligent response to water quality issues and non-compliances should they arise. Table 1 summarises the 2023/24 water quality localities.

A summary of the location of water quality localities in Barwon Water's service district is provided in Figure 2.

Table 1 Summary of water sampling localities

Water quality locality	Source water	Storage	Treatment plant	Number of Connections ¹	Permanent Population ²
Aireys Inlet	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	1,706	1,410
Anakie	Moorabool catchments Barwon catchments Ground water (drought)* Melbourne-Geelong Pipeline*	Stony Creek Reservoir Wurdee Boluc Reservoir	Moorabool Wurdee Boluc	302	650
Anglesea	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	3,253	3,130
Apollo Bay	Barham River	Marengo Basin	Apollo Bay	2,513	2,530
Bannockburn	Moorabool catchments**	Stony Creek Reservoir	Moorabool	2,590	7,210
Batesford	Moorabool catchments**	Stony Creek Reservoir	Moorabool	422	1,420
Bellarine	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	108	190
Birregurra	West Barwon Reservoir	Birregurra Basins	Birregurra	427	870
Clifton Springs	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	6,460	13,490
Colac	West Gellibrand and Olangolah Reservoirs	Colac basin no. 4 Colac basin no. 5	Colac	8,223	15,070
Cressy	West Gellibrand and Olangolah Reservoirs	Colac basin no. 4 Colac basin no. 5	Colac	207	310
Forrest	West Barwon Reservoir	Forrest Tank	Forrest	150	200

Table 1 continued

Water quality locality	Source water	Storage	Treatment plant	Number of Connections ¹	Permanent Population ²
Gellibrand	Lardners Creek	Gellibrand tanks	Gellibrand	73	300
Highton	Moorabool catchments	Stony Creek Reservoir Wurdee Boluc Reservoir	Moorabool	22,227	39,260
	Barwon catchments		Wurdee Boluc		
	Ground water (drought)*				
	Melbourne-Geelong Pipeline*				
Highton High Level	Barwon catchments	Wurdee Boluc Reservoir	Wurdee Boluc	1,208	3,370
	Ground water (drought)*				
Leopold	Barwon catchments	Wurdee Boluc Reservoir	Wurdee Boluc	7,993	19,320
	Ground water (drought)*				
Little River West	Moorabool catchments	Stony Creek Reservoir Wurdee Boluc Reservoir	Moorabool	154	390
	Barwon catchments		Wurdee Boluc		
	Ground water (drought)*				
	Melbourne-Geelong Pipeline*				
Lorne	Lorne catchment	Allen Reservoir	Lorne	2,458	2,000
Lovely Banks	Moorabool catchments	Stony Creek Reservoir Wurdee Boluc Reservoir	Moorabool	28,442	64,770
	Barwon catchments		Wurdee Boluc		
	Ground water (drought)*				
	Melbourne-Geelong Pipeline*				
Lovely Banks - Carrs Road	Moorabool catchments	Stony Creek Reservoir Wurdee Boluc Reservoir	Moorabool	46	100
	Barwon catchments		Wurdee Boluc		
	Ground water (drought)*				
	Melbourne-Geelong Pipeline*				
Montpellier	Moorabool catchments	Stony Creek Reservoir Wurdee Boluc Reservoir	Moorabool	29,435	68,050
	Barwon catchments		Wurdee Boluc		
	Ground water (drought)*				
	Melbourne-Geelong Pipeline*				
Moorabool	Moorabool catchments	Stony Creek Reservoir	Moorabool	696	1,910
Moriac	Barwon catchments	Wurdee Boluc Reservoir	Wurdee Boluc	348	970
	Ground water (drought)*				
Ocean Grove	Barwon catchments	Wurdee Boluc Reservoir	Wurdee Boluc	12,168	24,360
	Ground water (drought)*				
Pettavel	Barwon catchments	Wurdee Boluc Reservoir	Wurdee Boluc	22,632	58,080
	Ground water (drought)*				

Table 1 continued

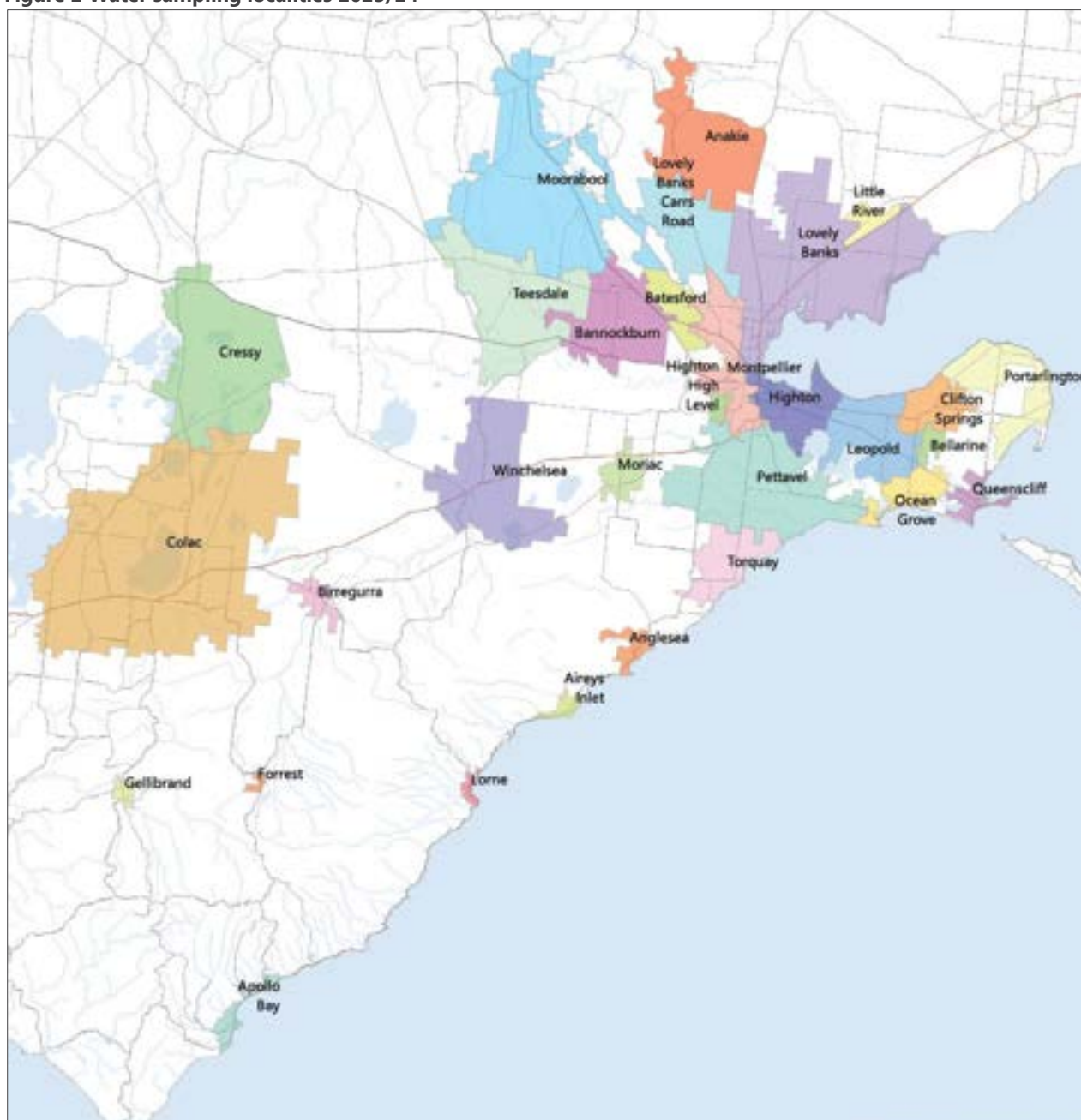
Water quality locality	Source water	Storage	Treatment plant	Number of Connections¹	Permanent Population²
Portarlinton	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	8,489	10,680
Queenscliff	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	4,631	5,820
Teesdale	Moorabool catchments**	Stony Creek Reservoir	Moorabool	1,531	4,810
Torquay	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	11,758	24,460
Winchelsea	Barwon catchments Ground water (drought)*	Wurdee Boluc Reservoir	Wurdee Boluc	1,267	2,710

1. This is the total number of properties connected to Barwon Water's waters supply system. This number includes residential and non-residential properties.
2. This is an estimated number of population serviced rounded to the closest 10. Each estimate is made by multiplying the number of residential connections by the occupancy rate (number of persons per dwelling) in ABS census data and adding any non-private dwellings (hospitals, nursing homes etc.).

* The Melbourne-Geelong Pipeline (MGP) and groundwater borefields are alternative water resources for improved water security. Water from the MGP was harvested for drinking water during the 2023/24 reporting period.

** These localities can also be fed from Wurdee Boluc WTP, however received water from Moorabool WTP only during the 2023/24 reporting period.

Figure 2 Water sampling localities 2023/24



4.3 Source of water

Drinking water is generally harvested from natural run-off into streams and reservoirs. Barwon Water's service area and supply system is relatively complex because it covers many small towns and water is harvested from a variety of different sources. All surface water catchments are declared Special Water Supply Catchments under the *Catchment and Land Protection Act 1994*.

Lorne, Apollo Bay, Gellibrand, Colac, Forrest and Birregurra are supplied from separate catchments all located in the Otway Ranges.

Barwon Water has a connection to Melbourne's water supply via the Melbourne to Geelong Pipeline. Barwon Water also holds a licence to extract water from the Anglesea borefield.

Torquay, Anglesea, Aireys Inlet, Winchelsea, Moriac, and the Bellarine Peninsula are all supplied from the Barwon catchment and can be supplemented by groundwater from the Anglesea borefield during times of drought. Geelong, Batesford, Anakie, Lara and Little River West are supplied by water from both the Moorabool and Barwon catchments.

Bannockburn, Teesdale, and Moorabool are supplied from the Moorabool catchment.

The Melbourne to Geelong Pipeline was also utilised to supplement drinking water in the Lovely Banks, Little River West, Montpellier and Highton localities in 2023/24.

Catchment management and protection is the first defence in the multi-barrier approach applied by Barwon Water to ensure safe drinking water. The condition of the catchment is arguably the most important factor influencing the quality of drinking water. It determines how much treatment is needed before the water is safe to drink and the costs associated with treatment. The effectiveness of catchment management is reliant on Barwon Water building and maintaining good relationships with stakeholders.

While most of our catchment areas are within protected national parks, some are open to the public and subject to a variety of land uses, including townships, rural living, forestry, agriculture, and recreation. Barwon Water has conducted comprehensive catchment risk assessments to assess the water quality risks associated with activities undertaken in the catchments. In addition, the impact of catchment activities on water quality is assessed through wide-ranging monitoring in catchments, streams, and reservoirs.

Barwon Water aims to prevent deterioration of water quality in our catchments in accordance with the priorities established in the Corangamite Regional Catchment Strategy, the Corangamite Waterway Strategy and the Victorian Waterway Management Strategy. We achieve this through several different strategies and initiatives prioritised within our Declared Special Water Supply Catchments, including:

- Planning and development controls
- Waterway protection and rehabilitation
- Fire management
- Prevention of soil erosion
- Community education
- Appropriate land management practices
- Restricting recreations activity
- Promoting awareness of the impact of human activity on our water quality.

4.4 Water treatment

After water is harvested from natural run-off into streams and reservoirs, or extracted from groundwater sources, it is filtered and disinfected. These processes provide key risk barriers to ensure safe drinking water. Filtration is the process of physically removing contaminants from drinking water. There are five types of filtration methods used at Barwon Water:

1. Dissolved air flotation filtration
2. Direct filtration
3. Conventional clarification/filtration
4. Micro filtration
5. Ultra filtration.

Disinfection is the process of inactivating pathogens so they cannot cause disease. Water disinfection is an important means of ensuring positive public health outcomes are achieved. The complex system of pipes used to distribute water means disinfection must be maintained from filtration through to our customers' taps, to prevent the regrowth of bacteria within the system. This is achieved using three main types of disinfection:

1. Chlorination
2. Chloramination
3. Ultraviolet light (UV).

The type of filtration and disinfection used depends on multiple factors including water quality, size of network and potential rate of consumption.

An overview of the treatment processes for each water sampling locality is provided in Table 2.

Over the 2023/24 reporting period there were no issues arising from the treatment plants.

Improvements to the treatment plants are explained in Section 9.

Table 2 Water filtration and treatment/disinfection processes

Water sampling locality	Main source	Water treatment plant	Treatment process	Treatment aids (Some aids only used seasonally)
Moorabool, Montpellier, Highton, Lovely Banks, Lovely Banks - Carrs Road, Bannockburn, Teesdale, Anakie, Little River West, Batesford.	East and West Moorabool River Barwon catchment (dependent on operational preference and time of year)	Moorabool	Dissolved Air Flotation Filtration (DAFF)	Aluminum Chlorohydrate Polymer
			Chlorination	Chlorine Gas
			Fluoridation	Fluorosilicic Acid
			Ultra Violet disinfection (supernatant return only)	
Lorne	Allen reservoir (St George River)	Lorne	Dissolved Air Flotation Filtration (DAFF)	Potassium Permanganate Aluminum Chlorohydrate Sodium Hydroxide Polymer Sodium Hexametaphosphate
			Chlorination	Sodium Hypochlorite
Colac, Cressy	West Gellibrand Reservoir & Olangolah Reservoir	Colac	Dissolved Air Flotation Filtration (DAFF)	Potassium Permanganate Soda Ash Aluminum Chlorohydrate Polymer Lime Sodium Hexametaphosphate
			Chlorination	Sodium Hypochlorite Chlorine Gas
			Fluoridation	Fluorosilicic Acid
Birregurra	West Barwon reservoir	Birregurra	Clarification	Aluminum Chlorohydrate
			pH Correction	Sulphuric Acid
			Microfiltration	*Sodium Hydroxide *Citric Acid (*used for cleaning membranes only)
			Chlorination	Sodium Hypochlorite

Table 2 continued

Water sampling locality	Main source	Water treatment plant	Treatment process	Treatment aids (Some aids only used seasonally)
Forrest	West Barwon Reservoir	Forrest	Clarification	Potassium Permanganate Sodium Hydroxide Polymer Aluminum Sulphate Sodium Hexametaphosphate
			Filtration	
			Chlorination	Sodium Hypochlorite
			UV disinfection	
Apollo Bay	Barham River (Marengo basin)	Apollo Bay	Dissolved Air Floatation (DAF) Clarification	Sulphuric Acid Aluminum Chlorohydrate
			Ultrafiltration	*Sodium Hydroxide *Citric Acid (*used for cleaning membranes only)
			Chlorination	Sodium Hypochlorite
			GAC (Granular Activated Carbon) filtration	
Aireys Inlet, Winchelsea, Moriac, Little River West, Highton High Level, Montpellier, Batesford, Pettavel, Highton, Lovely Banks, Lovely Banks - Carrs Road, Anakie, Torquay, Ocean Grove, Ocean Acres, Clifton Springs, Anglesea, Leopold, Portarlington, Queenscliff, Bellarine	West Barwon reservoir, East Barwon River, Callahan Creek diversion weir, Matthews Creek diversion weir, Pennyroyal Creek diversion weir and *Anglesea borefield	Wurdee Boluc *Pre-treatment for Anglesea borefield	Direct enhanced filtration	Aluminum Chlorohydrate Poly DADMAC Polymer
			Chlorination	Chlorine Gas
			Fluoridation	Fluorosilicic Acid
			*Aeration, Oxidation and pH adjustment	*Sodium hypochlorite *Lime *Potassium Permanganate *Polymer
Gellibrand	Lardners Creek	Gellibrand	Clarification	Aluminum Sulphate Polymer Soda Ash
			Filtration	
			Chlorination	Sodium Hypochlorite
			UV disinfection	

4.5 Distribution

After water is filtered and disinfected, it is conveyed to customers by a network of tanks, basins, pumps and pipes. Positive pressure and backflow prevention are the final barriers used by Barwon Water to ensure safe drinking water. Positive pressure means the water will direct any unwanted material away from the water supply system if leaks occur. Barwon Water requires the installation of testable backflow prevention devices when medium and high-hazard properties connect to the water supply system.

4.6 Secondary disinfection

Disinfectant residuals decrease as water passes throughout the distribution system and within storage tanks. Water disinfection plants are used to maintain adequate disinfection levels within the

distribution system. Barwon Water endeavours to maintain a balance between adequate disinfection residuals and aesthetic quality. Barwon Water manages 22 water disinfection plants. The two methods of secondary disinfection employed across the sites are chlorination and chloramination.

Chlorination is a widely used disinfection process, particularly effective against bacteriological organisms, and is most suited in shorter distribution systems.

Chloramination is an alternative disinfection process produced by combining chlorine with ammonia. The ammonia component stabilises the chlorine so disinfection can be sustained for longer in the drinking water distribution system.

Table 3 List of disinfection sites and disinfection mode

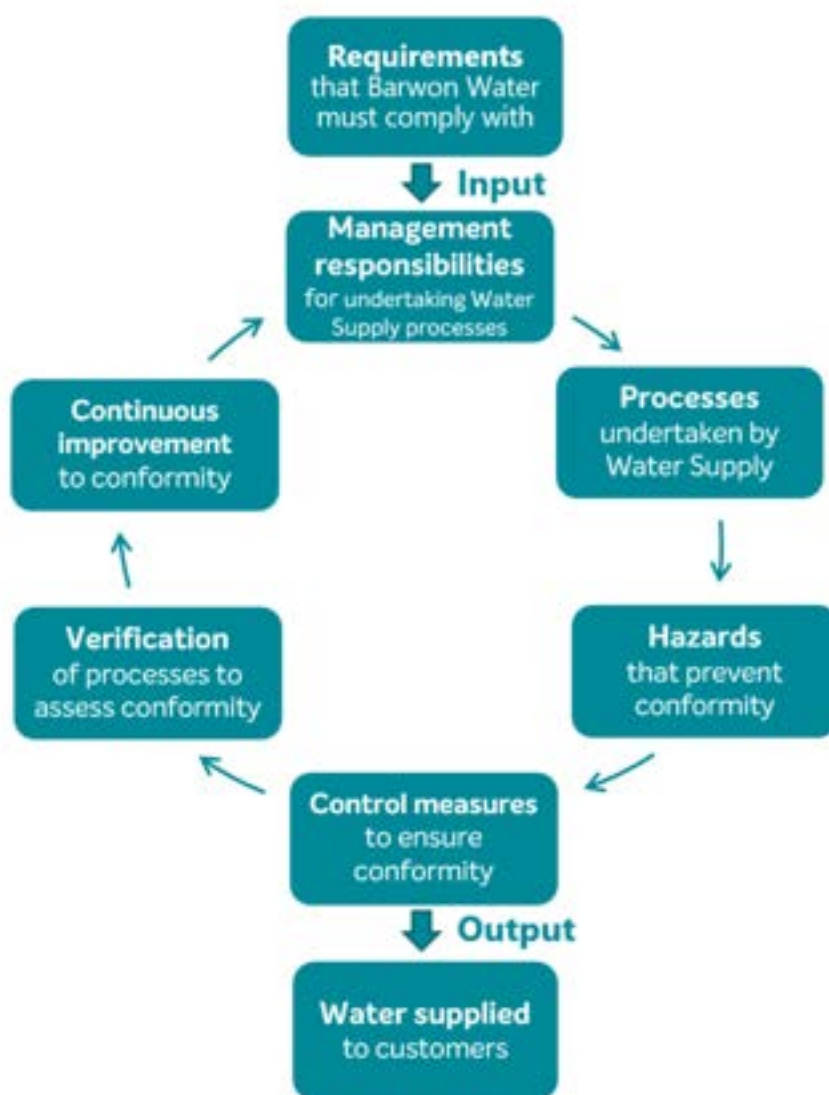
Water disinfection plant	Disinfection mode
Aireys Inlet	Chloramination
Alvie	Chlorination
Anakie	Chloramination
Anglesea	Chloramination
Bannockburn	Chloramination
Bellarine	Chlorination
Clifton Springs	Chloramination
Highton	Chlorination (seasonal operation)
Little River	Chlorination
Lorne	Chlorination
Lovely Banks 1,2&3	Chloramination
Lovely Banks – Carrs Rd	Chloramination
Meredith	Chlorination
Montpellier 2&4	Chlorination (seasonal operation)
Ocean Grove	Chloramination
Pettavel	Chlorination (seasonal operation)
Portarlington	Chloramination
Queenscliff	Chlorination
Teesdale	Chloramination
Torquay	Chlorination
Torquay High Level	Chlorination

5 Quality management system

5.1 Overview

Barwon Water utilises both AS/NZS ISO 9001:2015 and HACCP principles in its Quality Management System for the supply of safe and aesthetically acceptable drinking water. This system is summarised in Figure 3 below.

Figure 3 Quality Management System for the supply of drinking water



The requirements Barwon Water must meet, and the processes undertaken to meet these requirements, are described in previous sections of this report.

identification, control measures, verification and continual improvement.

The following pages outline other steps that form the Quality Management System, namely hazard

5.2 Hazard identification

To determine key risks to the supply of safe and aesthetically acceptable drinking water, an extensive hazard assessment process has been undertaken from catchment to customer connection involving independent experts, management, engineers and system operators.

The key risks resulting from this assessment are reviewed regularly to identify when new risks emerge or when existing risks change. In addition to system-wide assessments, each water treatment plant undergoes a thorough risk assessment based on asset/component and process failure.

5.3 Control measures

5.3.1 Critical control points

Control measures are established to reduce the key risks to the supply of drinking water to an acceptable level. Where control is vital, critical control points (CCPs) are established using HACCP principles. CCPs in the Barwon Water system include process steps such as chemical receipt, filtration, disinfection and fluoridation.

5.3.2 Maintenance programs

Maintenance programs are undertaken on Barwon Water assets to maintain and improve water quality. These include programmed mains flushing and air scouring, tank and basin cleaning, and risk-based treatment plant asset maintenance and replacement programs.

5.3.3 Trained operators

All operators involved in the water supply process undertake training to ensure a high knowledge of treatment processes and associated risks. Operators are required to achieve a minimum level of qualification and experience as per the National Water Industry Operator Certification Framework. Ongoing competency is managed and demonstrated through individual competency plans.

5.4 Verification and monitoring

Verification is used to authenticate the performance of the water supply system with respect to meeting process requirements. Verification at Barwon Water includes:

- Individual supervisory control and data acquisition systems (SCADA)
- System-wide telemetry providing monitoring and alarms 24 hours a day, 7 days a week
- On-call operators with remote access to the telemetry system
- A drinking water quality monitoring program conducted by an independent National Association of Testing Authorities (NATA) accredited laboratory with extensive monitoring at source, treatment and distribution
- A water treatment chemical assurance program that includes quality assured chemical suppliers, the provision of a certificate of analysis, and independent testing to verify the purity of the chemical
- Customer feedback captured in the complaints management system and customer perception surveys. These provide important information about the performance of the water supply system.

5.5 Continual improvement

Continuous improvement of water supply processes enables Barwon Water to continually comply with regulatory obligations and meet customer expectations. This is achieved through improvement actions identified from verification monitoring and through management review.

The production of an annual report directs Barwon Water to the key areas of improvement, particularly in relation to improvement projects. Section 9 of this report outlines some of the continuous improvement projects implemented in 2023/24 and maintains a strategic focus on continuously improving Barwon Water systems ongoing.

5.6 System auditing/peer review

An annual internal and external auditing program is established to assess the compliance of the quality management system.

5.7 Review of risk management plan

A review of the risk management plan was undertaken throughout the reporting period as part of Barwon Water's internal and external audit program.

The annual external audit of Barwon Water's quality management system was conducted by Intertek SAI Global. This was a surveillance audit for our HACCP system. The audit was successfully passed and provides confidence that we continually identify, assess, and manage drinking water quality risks.

There was no regulatory audit scheduled under the *Safe Drinking Water Act 2023* during 2023/24. All actions and opportunities for improvement from the previous regulatory audit in 2022/23 were completed and closed out during the previous reporting period.

6 Water quality for 2023/24

Barwon Water has developed a Drinking Water Quality Monitoring Program (DWQMP), undertaken by an independent laboratory, accredited by the National Association of Testing Authorities (NATA), under a laboratory services contract.

Source waters such as streams, channels and reservoirs, are analysed to provide early notice of changing upstream water quality and to verify catchment management plans are effective in minimising risks to source water quality.

Treated water is analysed to provide critical control point monitoring and to verify water quality risks associated with treatment plant operation are minimised.

Water throughout the distribution system and at designated customer water meters are analysed to verify Barwon Water is consistently delivering safe, high-quality water that meets relevant guidelines.

The DWQMP is based on sound risk management principles in accordance with Barwon Water's

HACCP plan and the recommendations of the Australian Drinking Water Guidelines.

The monitoring results and performance assessments of drinking water are presented in this report. Results provided in this report include all regulatory sampling conducted during the 2023/24 financial year.

Additional water quality data, including that which may prove helpful for enthusiasts such as home brewers can be found on the Barwon Water public website - barwonwater.vic.gov.au/water-and-waste/water-quality/results-and-data.

This data is updated quarterly to provide up-to-date, representative information to our customers.

If there is further data of interest, in compliance with Section 23 of the Act, all results from Barwon Water's drinking water quality monitoring program are available on request to the public within 7 days of being compiled.

6.1 Escherichia coli (*E. coli*)

- Limit of detection = 0 cfu/100 mL.
- Regulatory standard: All samples of drinking water collected contain no *E. coli* per 100 mL of drinking water, with the exception of any false positive sample.

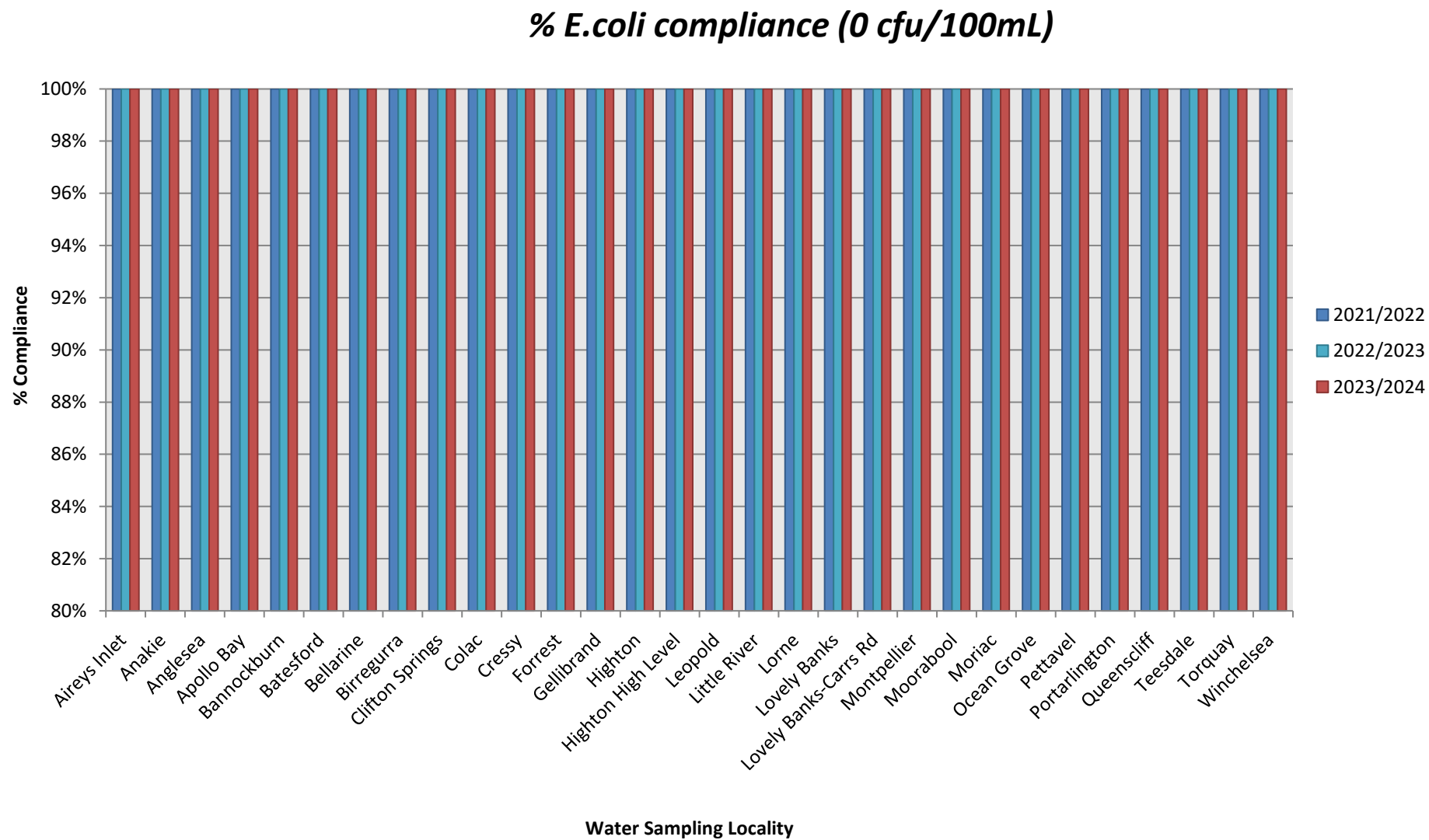
6.1.1 Results

Table 4 *E. coli* compliance

Water Quality Locality	Frequency of sampling*	Number of samples	Maximum Result (organisms/100 mL)	Number of detections and investigations conducted (s22)	Number of samples where standard was not met (s18)
Aireys Inlet	> weekly	76	0	0	0
Anakie	> weekly	76	0	0	0
Anglesea	> weekly	81	0	0	0
Apollo Bay	> weekly	86	0	0	0
Bannockburn	> weekly	75	0	0	0
Batesford	> weekly	75	0	0	0
Bellarine	> weekly	76	0	0	0
Birregurra	> weekly	75	0	0	0
Clifton Springs	> weekly	93	0	0	0
Colac	> weekly	201	0	0	0
Cressy	> weekly	75	0	0	0
Forrest	> weekly	75	0	0	0
Gellibrand	> weekly	75	0	0	0
Highton	> weekly	159	0	0	0
Highton High Level	> weekly	76	0	0	0
Leopold	> weekly	110	0	0	0
Little River	> weekly	76	0	0	0
Lorne	> weekly	91	0	0	0
Lovely Banks	> weekly	281	0	0	0
Lovely Banks-Carrs Rd	> weekly	76	0	0	0
Montpellier	> weekly	260	0	0	0
Moorabool	> weekly	99	0	0	0
Moriac	> weekly	74	0	0	0
Ocean Grove	> weekly	142	0	0	0
Pettavel	> weekly	205	0	0	0
Portarlinton	> weekly	107	0	0	0
Queenscliff	> weekly	86	0	0	0
Teesdale	> weekly	75	0	0	0
Torquay	> weekly	149	0	0	0
Winchelsea	> weekly	75	0	0	0

* Sampling frequency is dependent on population and varies in each locality as described in the Australian Drinking Water Guidelines 2011. This is in addition to distribution entry point samples taken to compare the representativeness of reticulation samples.

Figure 4 *E. coli*



6.1.2 Analysis of results

6.1.2.1 Trend over localities

In 2023/24 all of Barwon Water's water quality sampling localities complied with the *E. coli* water quality standard over the duration of the reporting period.

6.1.2.2 Trend over time

In 2023/24 compliance with the *E. coli* water quality standard was maintained at 100%.

6.2 Trihalomethanes (THMs)

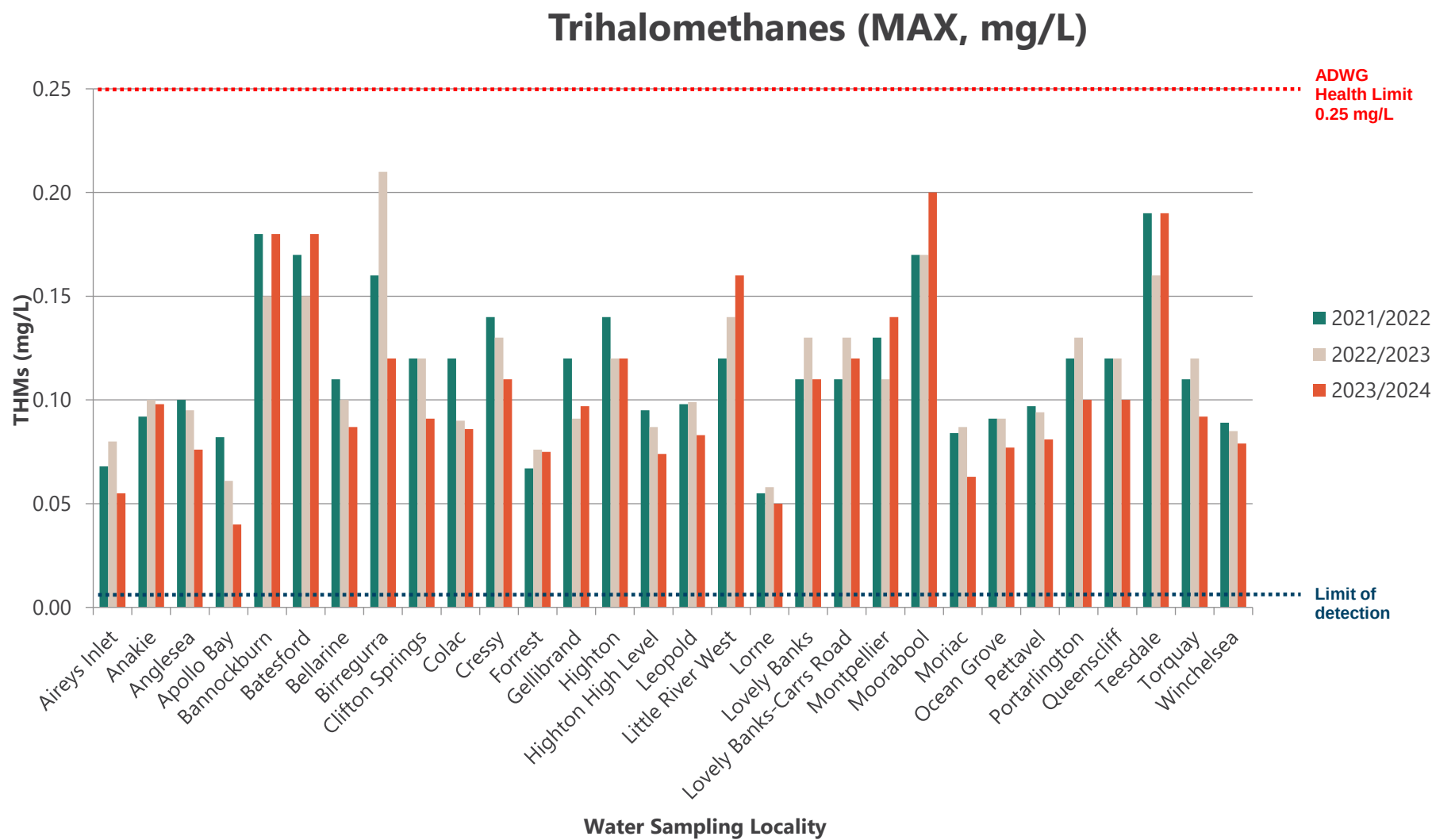
- Limit of detection = 0.004 mg/L
- Regulatory standard ≤ 0.250 mg/L

6.2.1 Results

Table 5 Trihalomethane results

Water Quality Locality	Frequency of sampling	Number of samples	Drinking Water Quality Standard (mg/L)	Maximum (mg/L)	Average (mg/L)	Number of samples where standard was not met (s18)
Aireys Inlet	> monthly	48	0.25	0.06	0.04	0
Anakie	> monthly	48	0.25	0.10	0.08	0
Anglesea	> monthly	48	0.25	0.08	0.07	0
Apollo Bay	> monthly	26	0.25	0.04	0.03	0
Bannockburn	> weekly	60	0.25	0.18	0.14	0
Batesford	> weekly	60	0.25	0.18	0.13	0
Bellarine	> monthly	48	0.25	0.09	0.07	0
Birregurra	> monthly	38	0.25	0.12	0.08	0
Clifton Springs	> monthly	48	0.25	0.09	0.08	0
Colac	> weekly	74	0.25	0.09	0.07	0
Cressy	> weekly	60	0.25	0.11	0.09	0
Forrest	> monthly	26	0.25	0.08	0.06	0
Gellibrand	> monthly	26	0.25	0.10	0.08	0
Highton	> monthly	48	0.25	0.12	0.08	0
Highton High Level	> monthly	48	0.25	0.07	0.06	0
Leopold	> monthly	48	0.25	0.08	0.07	0
Little River	> weekly	60	0.25	0.16	0.09	0
Lorne	> monthly	26	0.25	0.05	0.03	0
Lovely Banks	> weekly	118	0.25	0.11	0.08	0
Lovely Banks-Carrs Rd	> monthly	48	0.25	0.12	0.09	0
Montpellier	> weekly	72	0.25	0.14	0.09	0
Moorabool	> weekly	74	0.25	0.20	0.12	0
Moriac	> monthly	26	0.25	0.06	0.04	0
Ocean Grove	> monthly	48	0.25	0.08	0.06	0
Pettavel	> weekly	72	0.25	0.08	0.06	0
Portarlinton	> weekly	60	0.25	0.10	0.09	0
Queenscliff	> monthly	48	0.25	0.10	0.08	0
Teesdale	> weekly	60	0.25	0.19	0.13	0
Torquay	> weekly	72	0.25	0.09	0.07	0
Winchelsea	> monthly	48	0.25	0.08	0.06	0

Figure 5 Trihalomethanes



6.2.2 Analysis of results

6.2.2.1 Trend over localities

Full compliance was achieved across all water quality localities for Trihalomethanes (THMs). The highest result for the year was from the Moorabool water quality locality at 0.20 mg/L.

Elevated THMs results are common in water quality localities supplied by Moorabool Water Treatment Plant. This is due to the inherent characteristics of the supply catchment, seasonal higher level of organics in the source water and required chlorine disinfection dose.

6.2.2.2 Trend over time

Across most systems, maximum THMs levels in 2023/24 were slightly lower than in previous years. This can be associated with reduced levels of organics entering source water brought about by reduced rainfall.

Average THM levels followed a similar pattern to previous years.

6.3 Turbidity

- Limit of detection = 0.1 NTU
- Regulatory standard = The 95th percentile of results for samples in any 12-month period must be less than or equal to 5 NTU.

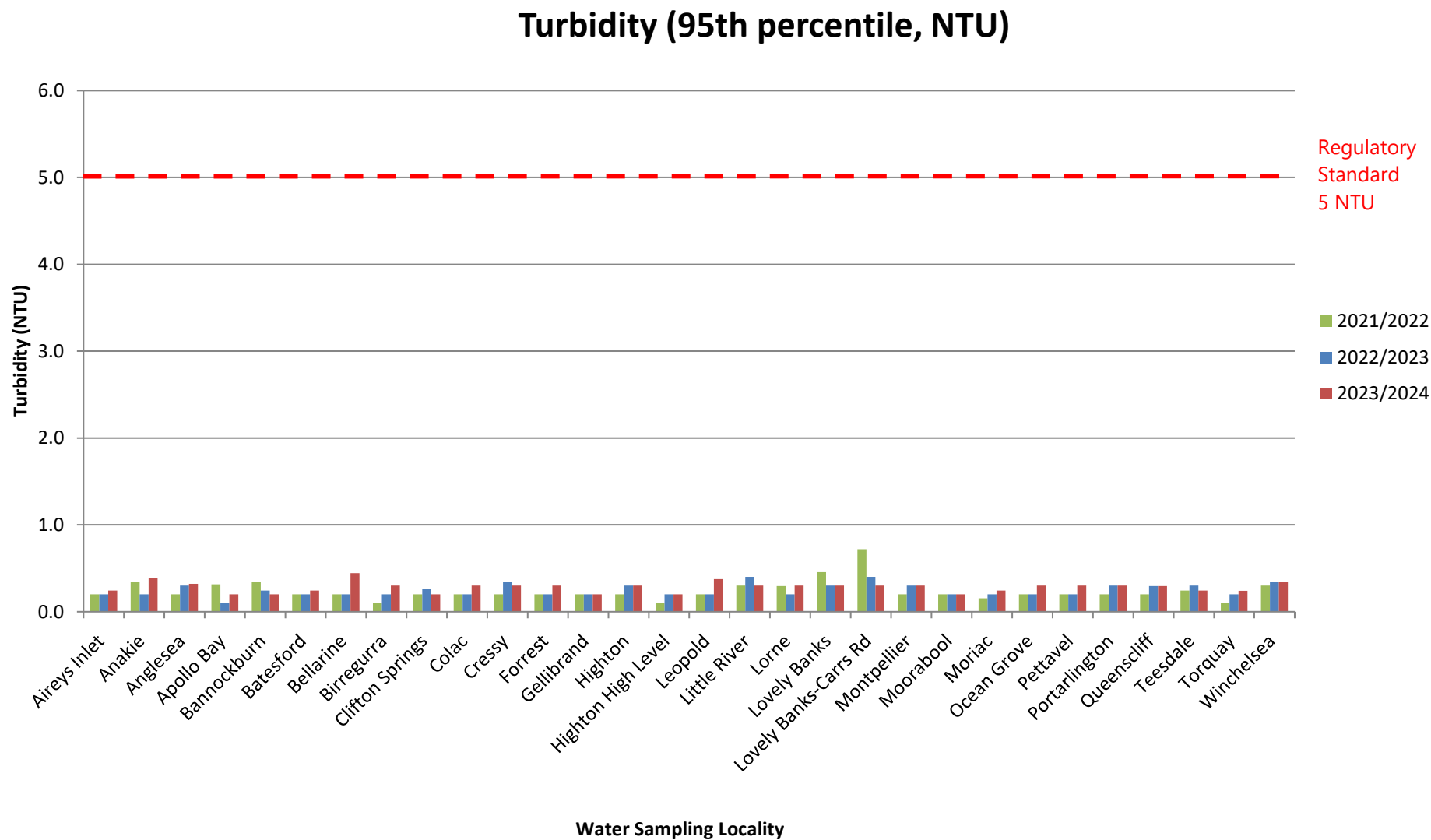
6.3.1 Results

Table 6 Turbidity

Water Quality Locality	Frequency of sampling*	Number of samples	Maximum (NTU)	Maximum 95th Percentile of turbidity results in any 12 months (NTU)	Number of 95th Percentile of results in any 12 months above the standard (s18)
Aireys Inlet	weekly	52	0.3	0.2	0
Anakie	weekly	52	0.5	0.4	0
Anglesea	> weekly	57	0.2	0.3	0
Apollo Bay	> weekly	62	0.4	0.2	0
Bannockburn	weekly	52	0.8	0.2	0
Batesford	weekly	52	0.6	0.2	0
Bellarine	weekly	52	8.7	0.4	0
Birregurra	weekly	52	0.4	0.3	0
Clifton Springs	> weekly	69	0.2	0.2	0
Colac	> weekly	74	0.3	0.3	0
Cressy	weekly	52	0.5	0.3	0
Forrest	weekly	52	0.6	0.3	0
Gellibrand	weekly	52	1.6	0.2	0
Highton	> weekly	135	3.3	0.3	0
Highton High Level	weekly	52	0.2	0.2	0
Leopold	> weekly	86	0.5	0.4	0
Little River	weekly	52	0.8	0.3	0
Lorne	> weekly	67	0.3	0.3	0
Lovely Banks	> weekly	187	1.0	0.3	0
Lovely Banks-Carrs Rd	weekly	52	7.5	0.3	0
Montpellier	> weekly	212	1.6	0.3	0
Moorabool	weekly	52	0.4	0.2	0
Moriac	weekly	52	0.5	0.2	0
Ocean Grove	> weekly	119	0.6	0.3	0
Pettavel	> weekly	157	7.1	0.3	0
Portarlinton	> weekly	83	0.7	0.3	0
Queenscliff	> weekly	62	0.5	0.3	0
Teesdale	weekly	52	0.5	0.2	0
Torquay	> weekly	113	0.4	0.2	0
Winchelsea	weekly	52	15.0	0.3	0

* Sampling frequency is dependent on population and varies in each locality as described in the Australian Drinking Water Guidelines 2011. This is in addition to distribution entry point samples taken to compare the representativeness of reticulation samples.

Figure 6 Turbidity



6.3.2 Analysis of results

6.3.2.1 Trend over localities

The turbidity (95th Percentile) standard was met in all localities during the reporting period.

6.3.2.2 Trend over time

Turbidity results during 2023/24 followed a similar pattern to previous years. All localities displayed results well below the quality standard.

6.4 Fluoride

The *Health (Fluoridation) Act 1973* states that the annual average fluoride concentration in drinking water shall not exceed 1 mg/L. The Department of Health (DH) directs that fluoride plants should achieve: a minimum average level greater than or equal to 0.6 mg/L (the lower action limit)

and an average level as close as possible to the operating target of 0.9 mg/L. The natural background fluoride levels from the catchment were monitored and are reported in Section 6.7.3 of this report.

6.4.1 Results

- Limit of detection = 0.05 mg/L
- ADWG health guideline value < 1.5 mg/L

Table 7 Fluoride

Water Quality Locality	Water Treatment Plant	Frequency of sampling	Number of samples	Drinking water quality standard (mg/L)	Target optimum fluoride concentration (mg/L)	Maximum (mg/L)	Average (mg/L)	Number of samples where standard was not met (s. 18)
Anakie	Moorabool WTP	monthly	12	1.5	0.9	0.88	0.80	0
Bannockburn		monthly	12	1.5	0.9	0.89	0.74	0
Batesford		monthly	12	1.5	0.9	0.94	0.76	0
Little River West		monthly	12	1.5	0.9	0.89	0.80	0
Lovely Banks		> monthly	42	1.5	0.9	0.88	0.72	0
Lovely Banks-Carrs Road		monthly	12	1.5	0.9	0.86	0.78	0
Montpellier		> quarterly	10	1.5	0.9	0.83	0.76	0
Moorabool		> monthly	62	1.5	0.9	0.90	0.72	0
Teesdale		monthly	12	1.5	0.9	0.92	0.75	0
Aireys Inlet	Wurdee Boluc WTP	monthly	12	1.5	0.9	0.88	0.79	0
Anglesea		monthly	12	1.5	0.9	0.87	0.76	0
Bellarine		monthly	12	1.5	0.9	0.88	0.75	0
Clifton Springs		monthly	12	1.5	0.9	0.89	0.80	0
Highton		> monthly	13	1.5	0.9	0.83	0.75	0
Highton HL		monthly	12	1.5	0.9	0.88	0.76	0
Leopold		monthly	12	1.5	0.9	0.85	0.75	0
Moriac		> weekly	65	1.5	0.9	0.94	0.72	0
Ocean Grove		monthly	12	1.5	0.9	0.89	0.79	0
Pettavel		monthly	12	1.5	0.9	0.86	0.75	0
Portarlinton		monthly	12	1.5	0.9	0.86	0.79	0
Queenscliff		monthly	12	1.5	0.9	0.84	0.77	0
Torquay		> quarterly	11	1.5	0.9	0.87	0.79	0
Winchelsea		monthly	12	1.5	0.9	0.88	0.77	0
Colac	Colac WTP	> weekly	101	1.5	0.9	0.91	0.71	0
Cressy		> weekly	52	1.5	0.9	0.92	0.75	0

Figure 7 Fluoride

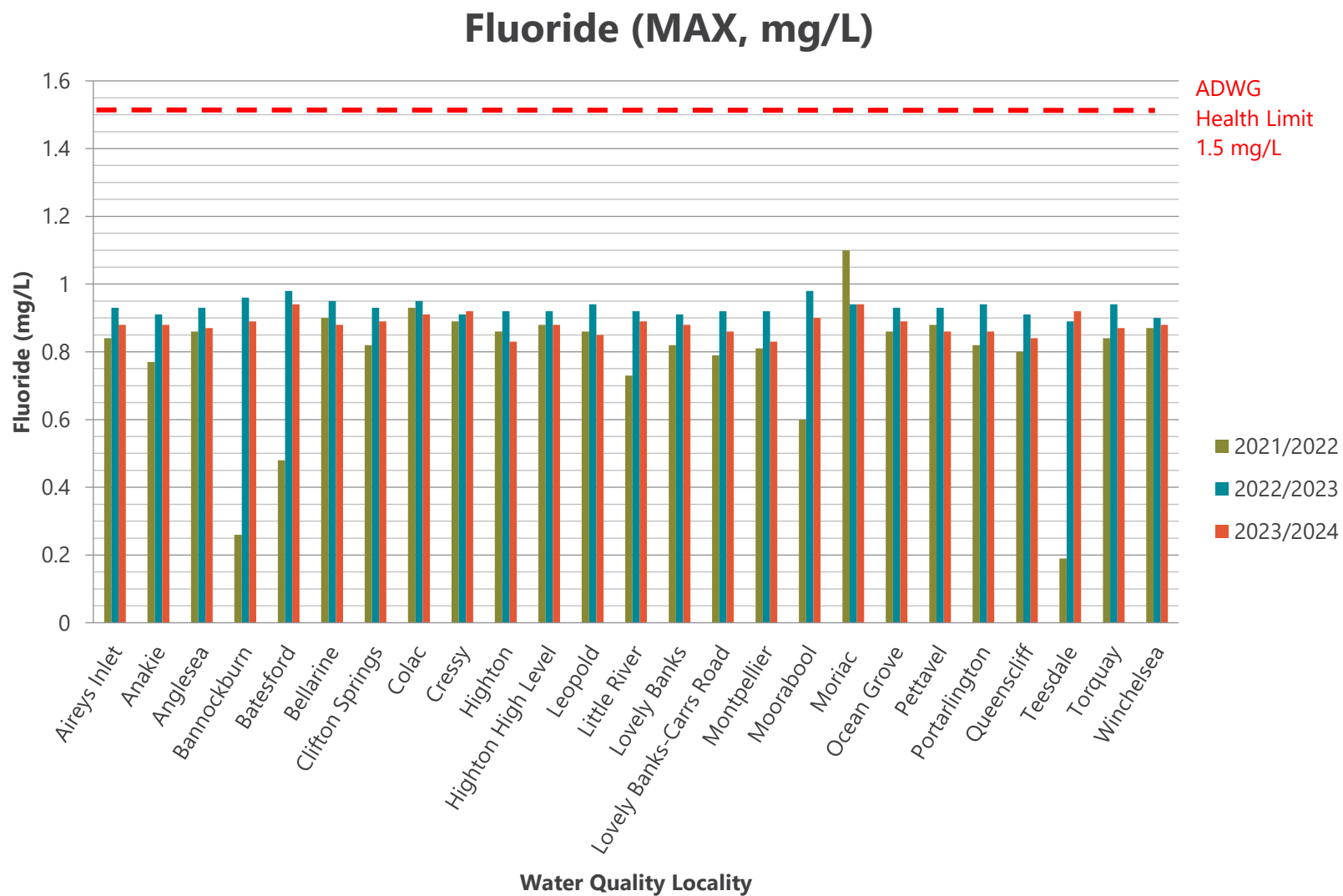
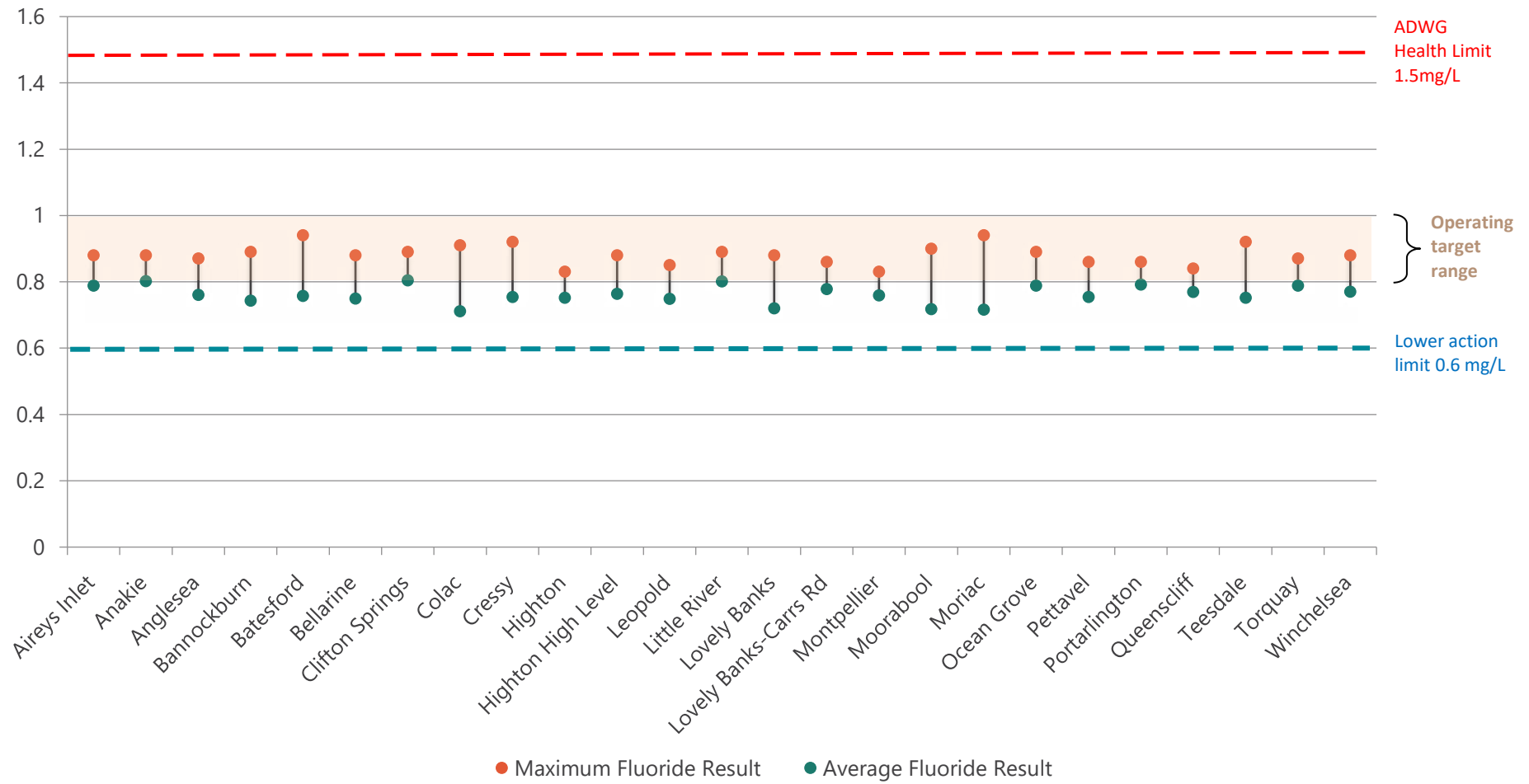


Figure 8 Fluoride (target)



6.4.2 Analysis of results

6.4.2.1 Trend over localities

The maximum fluoride level within all localities supplied by these treatment plants was below the drinking water quality standard value of 1.5 mg/L.

The fluoridation systems at Wurdee Boluc, Moorabool and Colac WTPs were upgraded during 2021/22. Since completion of the upgrades, all WTPs have been targeting the optimum dose of 0.9 mg/L.

In 2023/24, Barwon Water met the annual average rolling target of 0.8 mg/L at all treatment plants (measured via online instrumentation), as per the Fluoridation Code of Practice.

Fluoride dosing at Colac WTP was offline for approximately four days during the 2023/24 reporting period as notified to the Department of Health.

6.4.2.2 Trend over time

Average fluoride levels in the Colac and Cressy water quality localities improved in 2023/24. This was due to completion of optimisation works at the fluoridation plant to improve compliance with the rolling annual average target as specified in the Fluoridation Code of Practice.

Average fluoride levels across other fluoridated water quality localities were consistent with 2022/23 results.

6.5 Total chlorine

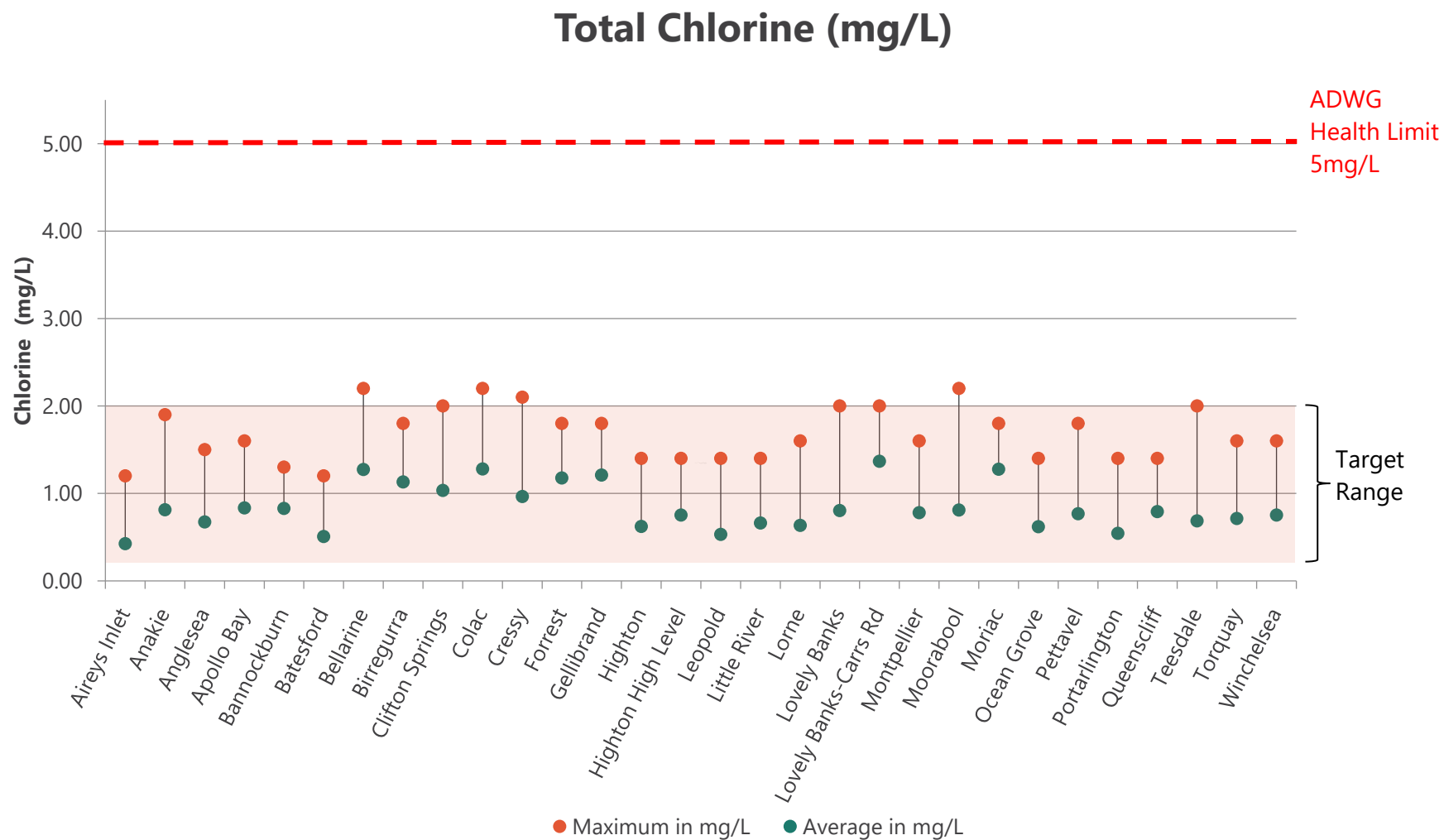
- Limit of detection = 0.1 mg/L
- ADWG health guideline value ≤ 5.0 mg/L

6.5.1 Results

Table 8 Total Chlorine

Water Quality Locality	Frequency of sampling	Number of samples	Maximum (mg/L)	Average (mg/L)	Complying (Yes/No)
Aireys Inlet	> weekly	76	1.20	0.42	Yes
Anakie	> weekly	76	1.90	0.81	Yes
Anglesea	> weekly	81	1.50	0.67	Yes
Apollo Bay	> weekly	86	1.60	0.83	Yes
Bannockburn	> weekly	75	1.30	0.83	Yes
Batesford	> weekly	75	1.20	0.51	Yes
Bellarine	> weekly	76	2.20	1.27	Yes
Birregurra	> weekly	75	1.80	1.13	Yes
Clifton Springs	> weekly	93	2.00	1.04	Yes
Colac	> weekly	201	2.20	1.28	Yes
Cressy	> weekly	75	2.10	0.96	Yes
Forrest	> weekly	75	1.80	1.18	Yes
Gellibrand	> weekly	75	1.80	1.21	Yes
Highton	> weekly	159	1.40	0.62	Yes
Highton High Level	> weekly	76	1.40	0.75	Yes
Leopold	> weekly	110	1.40	0.53	Yes
Little River	> weekly	76	1.40	0.66	Yes
Lorne	> weekly	91	1.60	0.63	Yes
Lovely Banks	> weekly	281	2.00	0.80	Yes
Lovely Banks-Carrs Rd	> weekly	76	2.00	1.37	Yes
Montpellier	> weekly	260	1.60	0.78	Yes
Moorabool	> weekly	99	2.20	0.81	Yes
Moriac	> weekly	74	1.80	1.28	Yes
Ocean Grove	> weekly	142	1.40	0.62	Yes
Pettavel	> weekly	205	1.80	0.77	Yes
Portarlinton	> weekly	107	1.40	0.54	Yes
Queenscliff	> weekly	86	1.40	0.79	Yes
Teesdale	> weekly	75	2.00	0.69	Yes
Torquay	> weekly	161	1.60	0.71	Yes
Winchelsea	> weekly	75	1.60	0.75	Yes

Figure 9 Total Chlorine (target)



6.5.2 Analysis of results

6.5.2.1 Trend over localities

During 2023/24, the maximum and average total chlorine results for most localities were within Barwon Water's target range. Results from individual localities were influenced by method of storage, system length, and water demand.

In all localities, the maximum total chlorine result was well below the ADWG health guideline value of 5.0 mg/L (4.1 mg/L in chloraminated systems).

All water quality localities met their target of maintaining minimum average total chlorine residual of >0.2 mg/L.

6.5.2.2 Trend over time

Average total chlorine results for the 2023/24 have seen a slight increase across most water quality localities compared against previous years.

This is primarily due to:

- An enhanced water mains preventative maintenance program which targets mains cleaning and flushing in poor performing areas of the network.
- An increased focus on end of system monitoring to achieve adequate chlorine residual, inform future water quality improvement plans and proactive system changes.

6.6 Analysis of results – summary for regulatory standards

Table 9 Compliance with regulatory standards outlined in Schedule 2 of the Safe Drinking Water Regulations (2015) over time – by analyte

Parameter	Percentage of water quality localities receiving compliant water										
	2013/ 14	2014/ 15	2015/ 16	2016/ 17	2017/ 18	2018/ 19	2019/ 20	2020/ 21	2021/ 22	2022/ 23	2023/ 24
<i>E. coli</i>	100	100	100	97	100	100	100	97	100	100	100
<i>Trihalomethanes</i>	100	100	100	100	100	100	100	100	100	100	100
<i>Turbidity</i>	100	100	100	100	100	100	100	100	100	100	100

In 2023/24, results were compliant with the drinking water quality standards (Schedule 2 of the Safe Drinking Water Regulations 2015) in all 30 water sampling localities.

Table 10 Compliance with regulatory standards outlined in Schedule 2 of the Safe Drinking Water Regulations (2015) over time – by percentage of population

Year	Percentage of population with complying water
2013/14	100
2014/15	100
2015/16	100
2016/17	99.63
2017/18	100
2018/19	100
2019/20	100
2020/21	99.99
2021/22	100
2022/23	100
2023/24	100

In 2023/24, 100 percent of the serviced population received water compliant with the drinking water quality standards.

6.7 Other health related parameters

Other drinking water quality standards under regulation 12(b) of the Safe Drinking Water Regulations 2015.

6.7.1 Arsenic

Barwon Water tests for the presence of arsenic in post-filtration water samples on a quarterly basis.

- Limit of detection = 0.001 mg/L
- ADWG health guideline value: <0.01 mg/L

Water Quality Locality	Source of Supply	Frequency of sampling	Number of samples	Maximum Result (mg/L)	Minimum Result (mg/L)	Complying (Yes/No)
Anakie	Wurdee Boluc WTP	Quarterly	4	<0.001	<0.001	Yes
Little River West						
Lovely Banks						
Lovely Banks-Carrs Road			4	<0.001	<0.001	Yes
Montpellier						
Highton						
Highton High Level	Wurdee Boluc WTP	Quarterly	4	<0.001	<0.001	Yes
Aireys Inlet						
Bellarine						
Clifton Springs						
Anglesea						
Ocean Grove						
Leopold						
Pettavel						
Portarlington						
Queenscliff						
Torquay						
Moriac						
Winchelsea						
Batesford	Moorabool WTP	Quarterly	4	<0.001	<0.001	Yes
Bannockburn						
Moorabool						
Teesdale						
Colac	Colac WTP	Quarterly	4	<0.001	<0.001	Yes
Cressy						
Apollo Bay	Apollo Bay WTP	Quarterly	4	<0.001	<0.001	Yes
Birregurra	Birregurra WTP	Quarterly	4	<0.001	<0.001	Yes
Forrest	Forrest WTP	Quarterly	4	<0.001	<0.001	Yes
Gellibrand	Gellibrand WTP	Quarterly	4	<0.001	<0.001	Yes
Lorne	Lorne WTP	Quarterly	4	<0.001	<0.001	Yes

Table 11 Arsenic

6.7.2 Copper

- Limit of detection = 0.001 mg/L
- ADWG health guideline value <2 mg/L

Table 12 Copper

Water Quality Locality	Frequency of sampling	Number of samples	Maximum Result (mg/L)	Average Result (mg/L)	Complying (Yes/No)
Aireys Inlet	monthly	12	0.019	0.005	Yes
Anakie	monthly	12	0.017	0.009	Yes
Anglesea	monthly	12	0.062	0.022	Yes
Apollo Bay	> monthly	16	0.010	0.007	Yes
Bannockburn	monthly	12	0.025	0.012	Yes
Batesford	monthly	12	0.036	0.014	Yes
Bellarine	monthly	12	0.040	0.017	Yes
Birregurra	> monthly	16	0.015	0.004	Yes
Clifton Springs	monthly	12	0.067	0.016	Yes
Colac	> monthly	16	0.009	0.004	Yes
Cressy	monthly	12	<0.001	<0.001	Yes
Forrest	> monthly	16	0.015	0.005	Yes
Gellibrand	> monthly	15	0.029	0.013	Yes
Highton	monthly	12	0.079	0.038	Yes
Highton High Level	monthly	12	0.023	0.013	Yes
Leopold	monthly	12	0.028	0.013	Yes
Little River	monthly	12	0.006	0.004	Yes
Lorne	> monthly	16	0.100	0.028	Yes
Lovely Banks	monthly	12	0.064	0.022	Yes
Lovely Banks-Carrs Rd	monthly	12	0.017	0.011	Yes
Montpellier	monthly	12	0.098	0.038	Yes
Moorabool	> monthly	15	0.082	0.018	Yes
Moriac	> monthly	16	0.022	0.009	Yes
Ocean Grove	monthly	12	0.065	0.022	Yes
Pettavel	monthly	12	0.099	0.021	Yes
Portarlinton	monthly	12	0.084	0.026	Yes
Queenscliff	monthly	12	0.200	0.064	Yes
Teesdale	monthly	12	0.038	0.015	Yes
Torquay	monthly	12	0.054	0.019	Yes
Winchelsea	monthly	12	0.110	0.028	Yes

6.7.3 Natural background fluoride

The following results are for the natural background fluoride levels from the catchment. The fluoride results for localities supplied with fluoridated water are reported in Table 7.

- Limit of detection = 0.05 mg/L
- ADWG health guideline value < 1.5 mg/L

Table 13 Natural background fluoride

Water Quality Locality	Source of supply	Minimum Frequency of sampling	Number of samples	Maximum (mg/L)	Minimum (mg/L)	Average (mg/L)
Anakie	Wurdee Boluc Reservoir,	Monthly	12	0.18	0.06	0.10
Little River West						
Lovely Banks	East and West Moorabool River		12	0.18	0.07	0.11
Lovely Banks-Carrs Road						
Highton						
Montpellier						
Bellarine	Wurdee Boluc Reservoir	Monthly	12	0.18	0.06	0.10
Clifton Springs						
Anglesea						
Ocean Grove						
Leopold						
Aireys Inlet						
Pettavel						
Portarlinton						
Queenscliff						
Torquay						
Moriac						
Winchelsea						
Highton High Level						
Batesford						
Bannockburn						
Moorabool						
Teesdale						
Colac	Colac Basin No. 4 & 5	Monthly	12	0.09	0.06	0.07
Cressy						
Apollo Bay	Marengo Basin	Monthly	12	0.12	0.07	0.09
Birregurra	Birregurra Basin	Monthly	12	0.16	0.07	0.13
Forrest	Forrest Basin	Monthly	12	0.09	0.05	0.07
Gellibrand	Lardners Creek	Monthly	12	0.10	0.06	0.08
Lorne	Allen Reservoir	Monthly	12	0.08	0.06	0.07

6.7.4 Lead

- Limit of detection = 0.001 mg/L
- ADWG health guideline value < 0.01 mg/L

Table 14 Lead

Water Quality Locality	Frequency of sampling	Number of samples	Maximum Result (mg/L)	Average Result (mg/L)	Complying (Yes/No)
Aireys Inlet	monthly	12	<0.001	<0.001	Yes
Anakie	monthly	12	<0.001	<0.001	Yes
Anglesea	monthly	12	0.002	0.001	Yes
Apollo Bay	> monthly	16	0.001	<0.001	Yes
Bannockburn	monthly	12	0.005	0.001	Yes
Batesford	monthly	12	0.001	<0.001	Yes
Bellarine	monthly	12	0.006	0.001	Yes
Birregurra	> monthly	16	<0.001	<0.001	Yes
Clifton Springs	monthly	12	<0.001	<0.001	Yes
Colac	> monthly	16	<0.001	<0.001	Yes
Cressy	monthly	12	<0.001	<0.001	Yes
Forrest	> monthly	16	<0.001	<0.001	Yes
Gellibrand	> monthly	15	<0.001	<0.001	Yes
Highton	monthly	12	0.001	<0.001	Yes
Highton High Level	monthly	12	<0.001	<0.001	Yes
Leopold	monthly	12	0.001	<0.001	Yes
Little River	monthly	12	<0.001	<0.001	Yes
Lorne	> monthly	16	0.004	0.001	Yes
Lovely Banks	monthly	12	0.002	0.001	Yes
Lovely Banks-Carrs Rd	monthly	12	0.001	<0.001	Yes
Montpellier	monthly	12	0.004	0.001	Yes
Moorabool	> monthly	15	0.001	<0.001	Yes
Moriac	> monthly	16	0.001	<0.001	Yes
Ocean Grove	monthly	12	<0.001	<0.001	Yes
Pettavel	monthly	12	<0.001	<0.001	Yes
Portarlinton	monthly	12	<0.001	<0.001	Yes
Queenscliff	monthly	12	0.007	0.002	Yes
Teesdale	monthly	12	<0.001	<0.001	Yes
Torquay	monthly	12	0.001	<0.001	Yes
Winchelsea	monthly	12	<0.001	<0.001	Yes

6.7.5 Manganese

- Limit of detection = 0.01 mg/L
- ADWG health guideline value < 0.5 mg/L

Table 15 Manganese

Water Quality Locality	Frequency of sampling	Number of samples	Maximum Result (mg/L)	Average Result (mg/L)	Complying (Yes/No)
Aireys Inlet	> monthly	24	<0.01	<0.01	Yes
Anakie	> monthly	24	<0.01	<0.01	Yes
Anglesea	> monthly	24	<0.01	<0.01	Yes
Apollo Bay	> monthly	48	<0.01	<0.01	Yes
Bannockburn	> monthly	24	<0.01	<0.01	Yes
Batesford	> monthly	24	<0.01	<0.01	Yes
Bellarine	> monthly	24	<0.01	<0.01	Yes
Birregurra	> monthly	47	<0.01	<0.01	Yes
Clifton Springs	> monthly	23	<0.01	<0.01	Yes
Colac	> monthly	47	0.020	<0.01	Yes
Cressy	> monthly	23	<0.01	<0.01	Yes
Forrest	> monthly	48	0.170	0.038	Yes
Gellibrand	> monthly	46	<0.01	<0.01	Yes
Highton	> monthly	24	<0.01	<0.01	Yes
Highton High Level	> monthly	24	<0.01	<0.01	Yes
Leopold	> monthly	24	<0.01	<0.01	Yes
Little River	> monthly	24	<0.01	<0.01	Yes
Lorne	> monthly	48	<0.01	<0.01	Yes
Lovely Banks	> monthly	24	<0.01	<0.01	Yes
Lovely Banks-Carrs Rd	> monthly	24	0.040	<0.01	Yes
Montpellier	> monthly	23	<0.01	<0.01	Yes
Moorabool	> monthly	47	<0.01	<0.01	Yes
Moriac	> monthly	46	<0.01	<0.01	Yes
Ocean Grove	> monthly	23	<0.01	<0.01	Yes
Pettavel	> monthly	23	<0.01	<0.01	Yes
Portarlinton	> monthly	24	<0.01	<0.01	Yes
Queenscliff	> monthly	24	<0.01	<0.01	Yes
Teesdale	> monthly	24	<0.01	<0.01	Yes
Torquay	> monthly	23	<0.01	<0.01	Yes
Winchelsea	> monthly	23	<0.01	<0.01	Yes

6.7.6 Other

The complete list of other health-related parameters analysed in 2023/24 are listed in Table 16, Table 17 and Table 18.

Table 16 Other health-related parameters for 2023/24

Analytes	Frequency of sampling	Health guideline compliance (Yes/No)
Antimony	Quarterly	Yes
Barium	Quarterly	Yes
Boron ¹	Quarterly	Yes
Cadmium	Quarterly	Yes
Chloral Hydrate (Trichloroacetaldehyde)	Annually	Yes
Chlorophenols ^{1 2}	Annually	Yes
Chlorine	Weekly	Yes
Chromium	Monthly	Yes
Cyanide ¹	Quarterly	Yes
Iodide ¹	Annually	Yes
Mercury	Annually	Yes
Molybdenum ¹	Annually	Yes
Monochloramine	Weekly	Yes
Nickel	Quarterly	Yes
Nitrate	Monthly	Yes
Nitrite	Monthly	Yes
Pesticides/herbicides ¹	Quarterly/monthly	Yes
Radioactive constituents	Annually	Yes
Selenium	Annually	Yes
Silver ¹	Annually	Yes
Sulphate	Quarterly/Monthly	Yes
Uranium ¹	Annually	Yes
Volatile organic carbons ¹	Annually	Yes

(1) Analytes only measured in source water pre-treatment. Measurement in source water is relevant to drinking water quality since it confirms that levels are below ADWG health guideline values prior to treatment.

(2) See Table 17 for a full list of pesticides/herbicides analysed

(3) See Table 18 for a full list of volatile organic carbons analysed.

All pesticides/herbicides are only monitored in the source water pre-treatment.

Table 17 Pesticides/herbicides analysed for in 2023/24 in source water

4,4-DDD	CHLOROPYRIFOS	2,4-DB
4,4-DDE	COUMPHANOS	2,4-DP
4,4-DDT	DEMETON-S	2,4-D
ALDRIN	DIAZINON	2,6-D
ATRAZINE	DICHLORVOS	4-Chlorophenoxyaceticacid
BHC (ALPHA ISOMER)	CHLORFENVINPHOS	Clopyralid
BHC (BETA ISOMER)	CHLOROTHALONIL	Dicamba
BHC (DELTA ISOMER)	EPN	MCPA
CIS-CHLORDANE	FENSULFONTHION	MCPB
DIQUAT	FENTHION	PHENOLS
DIELDRIN	MALATHION	TRICHLOPYR
ENDOSULPHAN I	METHYL PARATHION	HEXAZINONE
ENDOSULPHAN II	MONOCROTOPHOS	MOLINATE
ENDOSULPHAN	PARATHION	PICLORAM
ENDOSULPHAN SULPHATE	PHORATE	TEMEPHOS
ENDRIN ALDEHYDE	PROPHOS	PROPICONAZOLE
HEPTACHLOR EPOXIDE	RONNEL	PARAQUAT
HEPTACHLOR	TETRACHLORVINPHOS	SIMAZINE
HEXACHLOROBENZENE	TETRAETHYLDITHIOPYRPHOS	QUINTOZENE
LINDANE	TRICHLORINATE	PIRIMICARB
METHOXYCHLOR	TUKUTHION	DIURON
MECOPROP	2,4,5-T	PROPACHLOR
TRANS-CHLORDANE	2,4,5-TP	ENDRIN
GLYPHOSATE	2,4,6-T	AMITROL

All volatile organic compounds are monitored in both the source water pre-treatment, post-filtration water entry and reticulation.

Table 18 Volatile organic compounds analysed for in 2023/24

VINYL CHLORIDE	2,2-DICHLOROPROPANE	P-XYLENE
1,1,1,2-TETRACHLOROETHANE	2-CHLOROTOLUENE	NAPHTHALENE
1,1,1-TRICHLOROETHANE	4-CHLOROTOLUENE	N-BUTYLBENZENE
1,1,2,2-TETRACHLOROETHANE	BENZENE	N-PROPYLBENZENE
1,1,2-TRICHLOROETHANE	BROMOBENZENE	O-XYLENE
1,1-DICHLOROETHANE	BROMOCHLOROMETHANE	P-ISOPROPYLTOLUENE
1,1-DICHLOROETHENE	BROMODICHLOROMETHANE	SEC-BUTYLBENZENE
1,1-DICHLOROPROPENE	BROMOFORM	STYRENE
1,2,3-TRICHLOROBENZENE	CARBON DISULPHIDE	TERT-BUTYLBENZENE
1,2,3-TRICHLOROPROPANE	CARBON TETRACHLORIDE	TETRACHLOROETHENE
1,2,4-TRICHLOROBENZENE	CHLOROBENZENE	TOLUENE
1,2,4-TRIMETHYLBENZENE	CHLOROFORM	TRANS-1,2-DICHLOROETHENE
1,2-DIBROMO-3-CHLOROPROPANE	CIS-1,2-DICHLOROETHENE	TRANS-1,3-DICHLOROPROPENE
1,2-DIBROMOETHANE	CIS-1,2-DICHLOROPROPENE	TRICHLOROETHENE
1,2-DICHLOROBENZENE	DIBROMOCHLOROMETHANE	DICHLOROMETHANE
1,2-DICHLOROETHANE	DIBROMOMETHANE	BROMOMETHANE
1,2-DICHLOROPROPANE	ETHYLBENZENE	CHLOROETHANE
1,3,5-TRIMETHYLBENZENE	HEXACHLORO-1,3-BUTADIENE	DICHLORODIFLUOROMETHANE
1,3-DICHLOROBENZENE	ISOPROPYLBENZENE	TRICHLOROFLUOROMETHANE
1,3-DICHLOROPROPANE	METHYLENECHLORIDE	
1,4-DICHLOROBENZENE	M-XYLENE	

6.7.7 Analysis of results

6.7.7.1 Trend over localities

There were no non-compliances relating to other health-related water quality parameters, herbicides/pesticides or volatile organic compounds during 2023/24 in any locality. All results for these parameters were well below their respective health guideline value.

6.7.7.2 Trend over time

The results for these parameters in the 2023/24 period are consistent with those experienced in previous years.

6.8 Other aesthetic (not health-related)

6.8.1 True Colour

Based on aesthetic considerations, true colour in drinking water should not exceed 15 HU.

- Limit of detection = 1 Hu
- ADWG quality value < 15 Hu

Table 19 True Colour

Water Quality Locality	Frequency of sampling*	Number of samples	Maximum (HU)	Minimum (HU)	Average (HU)
Aireys Inlet	> weekly	76	3	<1	1.03
Anakie	> weekly	76	1	<1	<1
Anglesea	> weekly	81	1	<1	<1
Apollo Bay	> weekly	86	<1	<1	<1
Bannockburn	> weekly	75	3	<1	1.56
Batesford	> weekly	75	3	<1	1.69
Bellarine	> weekly	76	1	<1	<1
Birregurra	> weekly	75	1	<1	<1
Clifton Springs	> weekly	93	1	<1	<1
Colac	> weekly	201	1	<1	<1
Cressy	> weekly	75	<1	<1	<1
Forrest	> weekly	75	3	<1	1.23
Gellibrand	> weekly	75	1	<1	<1
Highton	> weekly	159	2	<1	1.01
Highton High Level	> weekly	76	2	<1	1.01
Leopold	> weekly	110	2	<1	1.01
Little River	> weekly	76	3	<1	1.08
Lorne	> weekly	91	1	<1	<1
Lovely Banks	> weekly	281	5	<1	1.05
Lovely Banks-Carrs Rd	> weekly	76	2	<1	1.04
Montpellier	> weekly	260	2	<1	1.01
Moorabool	> weekly	99	3	<1	1.57
Moriac	> weekly	74	1	<1	<1
Ocean Grove	> weekly	143	2	<1	1.01
Pettavel	> weekly	205	1	<1	<1
Portarlinton	> weekly	107	1	<1	<1
Queenscliff	> weekly	86	<1	<1	<1
Teesdale	> weekly	75	4	<1	1.77
Torquay	> weekly	161	1	<1	<1
Winchelsea	> weekly	75	3	<1	1.04

* Sampling frequency is dependent on population and varies in each locality, as described in the Australian Drinking Water Guidelines 2011.

6.8.2 Hardness (Total)

Total hardness is the sum of the concentrations of calcium and magnesium ions expressed as a calcium carbonate equivalent.

Hard water requires more soap than soft water to obtain lather. It can also cause scale to form on hot water pipes and fittings.

- Limit of detection = 1 mg/L
- ADWG quality value < 200 mg/L

Table 20 Hardness (total)

Water Quality Locality	Frequency of sampling*	Number of samples	Maximum (mg/L)	Minimum (mg/L)	Average (mg/L)
Aireys Inlet	monthly	12	52	34	41
Anakie	monthly	12	67	20	47
Anglesea	monthly	12	51	31	39
Apollo Bay	monthly	12	71	27	48
Bannockburn	monthly	12	130	55	91
Batesford	monthly	12	140	52	89
Bellarine	monthly	12	51	32	41
Birregurra	monthly	12	57	22	36
Clifton Springs	monthly	12	47	23	37
Colac	monthly	12	27	12	22
Cressy	monthly	12	27	18	23
Forrest	monthly	12	37	18	29
Gellibrand	monthly	12	85	18	44
Highton	monthly	12	61	20	40
Highton High Level	monthly	12	44	19	35
Leopold	monthly	12	41	23	36
Little River	monthly	12	78	17	48
Lorne	monthly	12	33	15	26
Lovely Banks	monthly	156	89	15	47
Lovely Banks-Carrs Rd	monthly	12	74	22	49
Montpellier	monthly	24	73	23	49
Moorabool	monthly	24	140	53	89
Moriac	monthly	12	53	26	37
Ocean Grove	monthly	12	47	30	38
Pettavel	monthly	24	51	18	36
Portarlinton	monthly	12	51	24	39
Queenscliff	monthly	12	44	23	35
Teesdale	monthly	12	140	59	92
Torquay	monthly	24	54	29	37
Winchelsea	monthly	12	38	19	34

6.8.3 pH

The pH value is a measure of hydrogen ion concentration in water. A pH value of 7.0 is neutral, values lower than 7.0 are acidic, and values higher than 7.0 are basic. The ADWG guideline range for pH is based on management of assets and is not related to health. According to current literature, a direct relationship between pH and human health is difficult to determine, as pH is closely associated with other aspects of water quality.

Consumption of food and beverages with quite low or high pH is common and does not result in adverse health effects (ADWG, 2011). A guideline value for pH between 6.5 and 8.5* is desirable to reduce corrosion and encrustation of pipes and fittings.

- Range of detection = 0 to 14
- ADWG value = 6.5 to 8.5

Table 21 pH

Water Quality Locality	Frequency of sampling	Number of samples	Maximum	Minimum	Average	Aesthetic operating range
Aireys Inlet	> weekly	76	8.3	6.9	7.4	6.5 - 8.5
Anakie	> weekly	76	7.8	6.8	7.3	6.5 - 8.5
Anglesea	> weekly	81	7.9	6.5	7.3	6.5 - 8.5
Apollo Bay	> weekly	86	7.8	6.7	7.3	6.5 - 8.5
Bannockburn	> weekly	75	7.9	6.6	7.2	6.5 - 8.5
Batesford	> weekly	75	7.9	6.6	7.2	6.5 - 8.5
Bellarine	> weekly	76	7.8	6.7	7.2	6.5 - 8.5
Birregurra	> weekly	75	8.2	6.4	7.5	6.5 - 8.5
Clifton Springs	> weekly	93	7.6	6.6	7.1	6.5 - 8.5
Colac	> weekly	201	7.8	6.2	7.0	6.5 - 8.5
Cressy	> weekly	75	8.8**	6.7	7.5	6.5 - 8.5
Forrest	> weekly	75	8.4	6.9	7.5	6.5 - 8.5
Gellibrand	> weekly	75	8.0	6.7	7.6	6.5 - 8.5
Highton	> weekly	159	7.5	6.6	7.1	6.5 - 8.5
Highton High Level	> weekly	76	7.5	6.5	7.1	6.5 - 8.5
Leopold	> weekly	110	7.5	6.6	7.0	6.5 - 8.5
Little River	> weekly	76	9.1*	6.8	7.5	6.5 - 8.5
Lorne	> weekly	91	8.2	6.6	7.2	6.5 - 8.5
Lovely Banks	> weekly	281	7.9	6.4	7.1	6.5 - 8.5
Lovely Banks-Carrs Rd	> weekly	76	7.8	6.6	7.2	6.5 - 8.5
Montpellier	> weekly	260	7.9	6.3	7.1	6.5 - 8.5
Moorabool	> weekly	99	8.2	6.6	7.2	6.5 - 8.5
Moriac	> weekly	74	7.8	6.6	7.0	6.5 - 8.5
Ocean Grove	> weekly	143	8.2	6.5	7.1	6.5 - 8.5
Pettavel	> weekly	205	7.4	6.6	7.0	6.5 - 8.5
Portarlington	> weekly	107	7.6	6.7	7.2	6.5 - 8.5
Queenscliff	> weekly	86	7.7	6.6	7.1	6.5 - 8.5
Teesdale	> weekly	75	9.0*	6.8	7.5	6.5 - 8.5
Torquay	> weekly	161	7.5	6.3	7.1	6.5 - 8.5
Winchelsea	> weekly	75	7.6	6.5	7.0	6.5 - 8.5

* Based on the need to reduce corrosion and encrustation in pipes and fittings, the pH of drinking water should be between 6.5 and 8.5. New concrete tanks and cement or mortar-lined pipes can significantly increase pH and a value above 8.5 may be tolerated, provided monitoring indicates no deterioration in microbiological quality. Barwon Water maintains a scheduled preventative maintenance program to proactively flush more vulnerable systems.

** Low turnover and cement and mortar-lined pipes contribute to some elevated pH results in Cressy. A scheduled maintenance program is in place to flush the network in Cressy. This mitigates the development of an elevated pH in the network. Additional weekly chlorine dosing is used to maintain residual in the network.

6.8.4 Conductivity

The electrical conductivity of water, measured in EC units, increases with the concentration of dissolved solids. A general rule of thumb is that electrical conductivity (EC) may be converted to total dissolved solids (TDS) by halving the value (an EC of 1000µS/cm is equivalent to a TDS of 500mg/L).

Based on taste, total dissolved solids in drinking water should not exceed 500 mg/L. No health effects have been associated specifically with high TDS concentrations (ADWG, 2011).

- Limit of detection = 1 µS/cm
- ADWG quality value < 1000 µS/cm

Table 22 Conductivity

Water Quality Locality	Source of supply	Frequency of sampling (min)	Number of samples	Maximum (µS/cm)	Minimum (µS/cm)	Average (µS/cm)
Anakie	Wurdee Boluc Reservoir,	twice monthly	24	300	220	235
Little River West						
Lovely Banks	East & West Moorabool River,	twice monthly	24	600	330	440
Lovely Banks-Carrs Road						
Montpellier	Melbourne-Geelong Pipeline	>twice monthly	48	140	76	110
Highton						
Aireys Inlet	Wurdee Boluc Reservoir	twice monthly	24	300	220	235
Bellarine						
Clifton Springs						
Anglesea						
Ocean Grove						
Leopold						
Pettavel						
Portarlington						
Queenscliff						
Torquay						
Moriac						
Winchelsea						
Highton High Level						
Batesford	East and West Moorabool River	twice monthly	24	600	330	440
Bannockburn						
Lethbridge						
Teesdale						
Colac	Colac Basins No. 4 & No. 5	twice monthly	24	160	130	142
Cressy						
Apollo Bay	Marengo Basin	twice monthly	24	300	240	268
Birregurra	Birregurra Basin	twice monthly	24	310	200	244
Forrest	Forrest Basin	twice monthly	24	400	150	314
Gellibrand	Lardners Creek	twice monthly	24	460	320	413
Lorne	Allen Reservoir	twice monthly	24	190	140	170

6.8.5 Analysis of results

6.8.5.1 Trend over localities

The average colour reported across all localities was at an acceptable level, with all localities recording a 1-2 HU colour average value well below the <15HU ADWG quality value.

The average pH reported in each locality was within the targeted 6.5-8.5 pH range. However, a number of samples from several localities reported maximum and minimum results outside the recommended pH range.

When these were identified, actions were implemented to correct pH levels within the network through increasing water turnover. The pH of the drinking water supply can become elevated due to natural leaching of calcium and hydroxide ions from cement linings in pipes and concrete in supply tanks.

All zones complied with the recommended hardness and conductivity levels.

6.8.5.2 Trend over time

The average colour, pH, hardness and conductivity, over each locality has followed a comparable pattern to previous years.

7 Emergency/ incident management

7.1 Emergencies/events –

Section 22 notifications

There were two Section 22 notifications made to the Department of Health during the 2023/24 reporting period, as summarised in Table 23 below.

Table 23: Section 22 Notifications 2023/24

Date	Type	Water Quality Locality	Nature of s22 Notification	Corrective Actions Taken	Preventative Actions Taken
13 Oct 2023	Incorrect connection of drinking water service line to Class A recycled water main	Pettavel	Following a dirty water complaint, Barwon Water identified a colour discrepancy between the drinking water and recycled water taps at the property's meter. An investigation confirmed that the single dwelling's drinking water service line had been incorrectly connected to the Class A recycled water main. This error occurred during construction by the developer's contractors.	Rectified the incorrect connection so that the drinking water service line is connected to the drinking water main and verified water quality through sampling	Letter sent to all developer's consultants and contractors on importance of correct connection and associated obligations and responsibilities with respect to relevant processes. Put in place additional auditing of developer's contractors when constructing dual pipe assets. Put in place additional downstream tests to detect this issue in the next phases of construction, prior to commissioning.
21 May 2024	Incorrect connection of potable service line to Class A recycled water main	Pettavel	Following a taste & odour complaint, Barwon Water investigated and identified a discrepancy in the Total Dissolved Solids (TDS) measured at the drinking water and recycled water taps at the property's meter. The investigation confirmed that the single dwelling's drinking water service line had been incorrectly connected to the Class A recycled water main. This error occurred during construction by the developer's contractors.	Rectified the incorrect connection so that the drinking water service line is connected to the drinking water main and verified water quality through sampling	Engaged directly with the developer's consultant and contractor, on further uplifts to their quality assurance processes. (The relevant assets were constructed prior to those described in the row above, so action had already been otherwise taken to reduce future likelihood of reoccurrence).

7.2 Emergencies/events – Section 18 notifications

No Section 18 notifications were made during the 2023/24 reporting period.

8 Water Quality Complaints

8.1 Complaints 2023/2024

In accordance with our commitment to providing high quality drinking water, complaints regarding drinking water quality are taken very seriously. Barwon Water records all complaints relating to drinking water quality.

The complaints are compared to the number of connected properties to enable comparison with other water suppliers. The number of complaints in 2023/2024 was 1.47 per 1,000 properties. This is a slight increase compared to last year's complaints trend (1.17 complaints per 1,000 properties), however is consistent with the 3-year rolling average.

The largest complaints category was for Taste and Odour. This can be associated with operational changes to balance water resources due to raw water challenges created by a blue green algae bloom at Wurdee Boluc reservoir. These operational changes resulted in some customers experiencing taste profile changes. Customers were kept informed via a proactive communication campaign.

Although there was an increase in discoloured water and other complaints. These were not attributed to specific events, and many were associated with corrosion of customer plumbing beyond the point of supply (the water meter).

White water complaints also increased, and where possible, opportunities to reduce these have been sought.

Table 24 Comparison of complaints for financial years 2023/24 and 2022/23

Type of complaints	Number of complaints			Comparison to 2022/23 reporting period	Comments
	2023/24	2022/23	2021/22		
Alleged Diagnosed Illness	0	0	0	0	Comparable figure to previous year.
Discoloured Water	88	80	66	+8	10% increase in number of complaints compared to previous year.
Taste or Odour	91	62	70	+29	47% increase in number of complaints compared to previous year.
White Water	36	19	26	+17	89% increase in number of complaints compared to previous year.
Other	52	48	39	+4	8% increase in number of complaints compared to previous year.
TOTAL	267	209	201	+58	28% increase in number of complaints compared with previous year (see Figure 10).

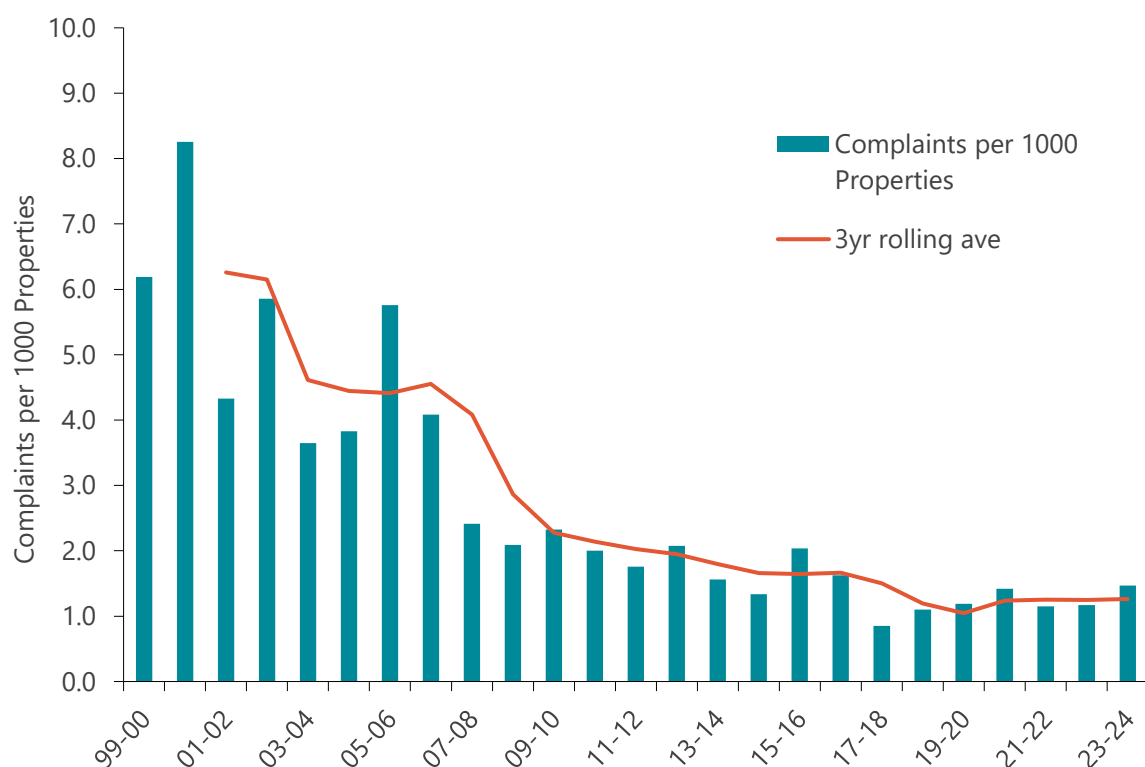
Table 25 Customer complaints summary for each sampling locality 2023/24

Water Quality Locality	Type of complaint					Total
	Alleged Diagnosed Illness	Discoloured Water	Taste or odour	White water	Other	Total
Aireys Inlet	0	1	2	0	0	3
Anakie	0	0	0	0	0	0
Anglesea	0	0	1	2	1	4
Apollo Bay	0	0	2	0	0	2
Bannockburn	0	1	0	2	1	4
Batesford	0	3	1	0	0	4
Bellarine	0	0	0	0	0	0
Birregurra	0	0	0	0	1	1
Clifton Springs	0	5	2	0	4	11
Colac	0	9	2	2	3	16
Cressy	0	0	0	0	0	0
Forrest	0	4	1	0	1	6
Gellibrand	0	0	0	0	0	0
Highton	0	12	21	5	10	48
Highton HL	0	0	0	0	0	0
Leopold	0	1	2	3	0	6
Little River West	0	0	0	4	0	4
Lorne	0	2	1	1	0	4
Lovely Banks	0	13	9	2	4	28
Lovely Banks-Carrs Road	0	0	0	0	0	0
Montpellier	0	12	14	0	12	38
Moorabool	0	3	0	0	0	3
Moriac	0	0	0	0	0	0
Ocean Grove	0	3	1	0	2	6
Pettavel	0	14	12	10	7	43
Portarlinton	0	0	5	2	1	8
Queenscliff	0	0	1	1	0	2
Teesdale	0	1	3	2	1	7
Torquay	0	4	8	0	1	13
Unknown locality - no address provided	0	0	3	0	3	6
Winchelsea	0	0	0	0	0	0
TOTAL	0	88	91	36	52	267

Table 26 Table of customer complaints by type for 2023/24

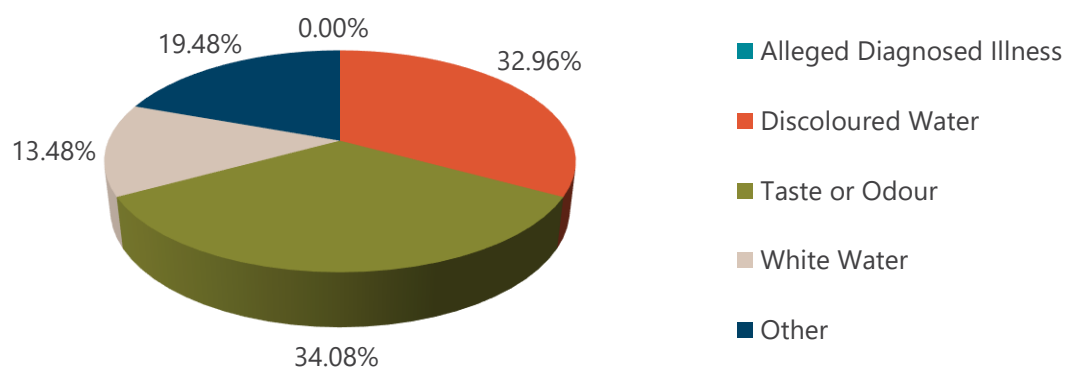
Type of Complaint	Number of Complaints 2023/24	Percentage of Total Complaints	Number of complaints per 1000 properties 2023/24
Alleged Diagnosed Illness	0	0%	0.00
Discoloured Water	88	33%	0.48
Taste or Odour	91	34%	0.50
White Water	36	13%	0.20
Other	52	19%	0.29
TOTAL	267	100%	1.47

Figure 10 Complaints per 1000 properties 2023/24



Most complaints for 2023/24 related to taste or odour and odour and discoloured water (Figure 11). Table 25 shows a breakdown of complaints by complaint type. This is a similar breakdown to that recorded in previous years.

Figure 11 Customer complaint summary by type 2023/24



8.2 Complaint response

Customer complaints are mostly received by Barwon Water's Customer Contact Centre and registered on our customer and complaints management system. Barwon Water also occasionally receives complaints via social media, email, or the enquiry function of the public website. This feedback is also registered in the complaints and customer management system and managed in the same way as telephone complaints.

During business hours, the Product Quality team endeavour to respond as quickly as possible (within the same day) to liaise with the customer to resolve the problem. Outside business hours, the Duty Officer will respond to complaints and initiate corrective action immediately where required. These complaints are then assessed to determine if follow up contact is required by a member of the Product Quality team during the next business day.

8.2.1 Discoloured water

Discoloured water can be caused by multiple different factors, including burst main events, manganese and iron oxidation and problems which may relate to the internal plumbing at a customer's property. The response given to these complaints varies depending on the nature of the cause, but may include reactive mains cleaning, reviewing treatment performance or advising the customer to have their internal plumbing inspected by a licensed plumber.

If a customer reports blue discolouration, this is usually caused by copper corrosion. When investigated, the issue is often traced back to the internal plumbing on the customer's property as Barwon Water does not use copper pipes or fittings. Advice is usually provided to customers to engage a licensed plumber to make an assessment.

8.2.2 Taste/odour

Taste and odour issues are usually related to chlorine in the water or a change in the raw water quality, such as an increase in organics or algae in the raw water. In responding to these complaints, treatment processes and recent monitoring results are reviewed and, where possible, alterations are made. In cases where a customer may be particularly sensitive to taste and odour, advice is given on methods to address the issue.

8.2.3 White water

When air becomes trapped in water supply, it has a white or "milky" appearance. This is generally caused by repairs/replacements of water mains. Where possible, attempts are made to remove the affected water from the reticulation system by flushing the main at the location where air is gathering. However, as this is not always possible, customers may be asked to wait for the air to be removed naturally, which can take two to three days. Customers are advised that the appearance of the water is caused by trapped air becoming entrained in the water supply when it is under pressure in the main.

When a white-water complaint is received a number of factors are evaluated to determine if flushing is necessary including:

- verifying the source to ensure the water supply remains safe,
- topography of the area (to determine where the air is likely to be accumulating in the main),
- number of connections in the area (likelihood that the air will be flushed out via normal use),
- how long the problem has been occurring for,
- any other factors that may affect the customer.

If necessary, a flush will be arranged at the time of complaint. Alternatively the customer may be advised that the problem should clear naturally and to call back if it is persisting after 1-2 days. If a second call is received, a flush is arranged.

8.2.4 Alleged diagnosed illness

Illness complaints are generally received from customers who wish to confirm whether the water they have been drinking may have made them ill. To address this type of complaint a review of monitoring results is undertaken. In some cases, independent water quality testing at the customer's meter may also be arranged.

If there is no evidence indicating the delivered water has been unsafe, the customer is advised to consult a doctor if they have further concerns.

A standard operating procedure has been developed by Barwon Water which is aligned with Department of Health (DH) guidance. The procedure has been adopted to ensure all illness complaints prompt a thorough investigation of the water supply system to confirm there have been no system failures which could provide an opportunity for unsafe water to be supplied.

If at any point in time Barwon Water believes there has been potential for unsafe water to be supplied, DH will be notified (via a Section 22 notification).

If multiple illness complaints are received from the same water quality locality within a 7-day period, and investigations show there has been no potential for the water supply to be the cause, Barwon Water will notify DH as a courtesy to enable them to investigate further.

9 Water quality improvement projects

9.1 Forrest and Gellibrand WTP upgrade project

In 2023/24 construction was completed on the upgraded Forrest Water Treatment Plant, replacing the previous plant which was approaching the end of its service life.

Construction has also commenced on the upgraded Gellibrand WTP, with the new plant expected to be completed in 2024/25.

Both upgraded plants are within existing site footprints and have been designed to increase resilience against variable raw water quality.

9.2 Completion of Lorne WDP

In 2023/24 we constructed and commissioned a new secondary disinfection plant supplying Lorne Tank No.3. The project will improve water quality for the Lorne area by providing reliable and consistent chlorine residual through the network.

9.3 Installation of aerators and ultrasonic units in Wurdee Boluc Reservoir

In 2023/24 we installed aerators and two ultrasonic units in Wurdee Boluc Reservoir. The aeration and ultrasonic units will improve raw water quality supplied to Wurdee Boluc WTP by improving mixing and disrupting the growth potential of blue-green algae in the reservoir.

9.4 Catchment management and river health

Barwon Water harvests water from declared drinking water supply catchments on the Barham, Upper Barwon, Gellibrand, Moorabool and St George Rivers and their tributaries. While these are largely contained within publicly managed national parks and reserves, significant areas of private land are also present within some catchments.

We have a keen interest in ensuring the protection and rehabilitation of land and waterways within these catchments. This is undertaken in partnership with key stakeholders and in accordance with the priorities established in the Corangamite Regional Catchment Strategy, the Corangamite Waterway Strategy and the Victorian Waterway Management Strategy.

In 2023/24, we invested more than \$2.6M to protect and improve catchment and waterway health. This included continuation of our ongoing partnership with the Corangamite Catchment Management Authority through our River Health Agreement and the Living Moorabool and Upper Barwon Flagship Projects. We also continued our long-standing partnerships with the Moorabool Catchment Landcare Group, Southern Otway Landcare Network and Upper Barwon Landcare Network to support a coordinated approach to deliver on-ground protection and improvement works within drinking water supply catchments.

As part of these works, we also completed on-ground works for Stage 2 of a \$4.3 million project to

remove invasive willows, stabilise stream banks, and revegetate streamside buffers along a 4 km reach of the East Barwon River. This is a significant project and is critical for improving flows in the river and reducing waterlogging of adjacent land. Stock exclusion fencing and revegetation will also improve and protect water quality.

In collaboration with the Corangamite CMA and Landcare, we also continued ongoing maintenance of large river restoration projects on Dewing Creek (a tributary of the Barwon River), and the East Moorabool River at Bolwarra Weir. These projects included willow removal, fencing for stock exclusion and revegetation to improve water quality.

In collaboration with the Conservation Ecology Centre and Parks Victoria, we also monitored and supported the management of feral pig and deer populations at a number of locations within drinking water supply catchments.

In 2023/24 we also continued our support for the ongoing operation of the Corangamite Citizen Science program. This program supports community groups to conduct water quality testing at monitoring sites and delivers education programs for students in the Corangamite region. Barwon Water staff have also been involved in presentations and public events promoting river health and catchment issues. The program also includes water quality testing at Waterwatch sites selected and monitored by the Wadawurrung Traditional Owners Aboriginal Corporation, with results published in a water quality report produced in partnership with Wadawurrung Traditional Owners.



 BarwonWater

Appendices

Appendix A – Current HACCP Certification



BUREAU VERITAS
Certification

Certification
Awarded to

BARWON REGION WATER CORPORATION
61-67 RYRIE STREET, GEELONG, VIC, 3220
AUSTRALIA

Bureau Veritas certify that the Management System of the above organisation has been audited and found to be in accordance with the requirements of the management system standards indicated below

STANDARD

HACCP as defined by
CAC/RCP 1-1969, REV4, 2003

SCOPE OF SUPPLY

THE REGISTRATION COVERS THE FOOD SAFETY MANAGEMENT SYSTEM FOR THE HARVESTING, TREATMENT AND DISTRIBUTION OF DRINKING WATER TO A BROAD RANGE OF CUSTOMERS

Original Approval Date: **16 July 2004**
Subject to the continued satisfactory operation of the organisation's Management System,
this certificate is valid until: **13 November 2024**

To check the validity of this certificate please call tel: **1800 855 190**
Further clarification regarding the scope of this certificate and the applicability of the Management System requirements may be obtained by consulting the organisation.

Certificate Number: **AU004804-1**  Date: **13 November 2021**

Andrew Mortimore
Vice President – IAF Pacific Region

Managing office: Bureau Veritas Pty Ltd, 3/435 Williamstown Road,
Port Melbourne, Victoria, 3207
Issuing office: Bureau Veritas Pty Ltd, 3/435 Williamstown Road,
Port Melbourne, Victoria, 3207



Appendix B – 2022/23 Safe Drinking Water Audit Certificate & Correction Letter



ABN 43 143 282 399 ☎ +61 411 048 544 ✉ contact@riskedge.com.au www.riskedge.com.au

Risk Management Plan Audit Certificate

Safe Drinking Water Regulations 2015

Certificate Number:	BW-346
Auditor Notice of Approval Conditions #:	181
Audit period:	1 January 2021 to 31 December 2022
To:	Ms Komal Dalal Product Quality Lead Asset, Systems & Environment Barwon Water 55-67 Ryrie Street (PO Box 659) Geelong VIC 3220
Australian Business Number (ABN):	86 348 316 514

I, Dr Annette Davison, after conducting a risk management plan audit of the water supplied by Barwon Water, am of the opinion that Barwon Water has not complied with the obligations imposed by section 8(1) of the *Safe Drinking Water Act 2003* during the audit period.

The details of the reasons for noncompliance (all minor) are detailed in the table below.

Signed:

Date: 2023-04-22

Auditable Area	Noncompliances	OFIs
1 Activities Conducted to Manage Risk	The Act requires development and implementation of preventative strategies (including appropriate control and monitoring measures). (s. 9(1)(d)). While minor, a GAC Filtration CCP was identified as part of the Apollo Bay WTP Process Control, but not listed on the CCP Locations list, creating potential for a gap in process understanding. OFIs (OFI 1D-1 and OFI 1D-3) have been captured to address this gap. The Act requires implementation and compliance with the requirements of the risk management plan (s. 7(1)(b), s. 8(1)(b)). Some minor noncompliances in implementation were observed including gaps in completion of chemical deliveries record requirements, omission of one test on PAC23 (Colac WTP), raw and treated water turbidity meters at Colac WTP not been externally calibrated when due and instrument numbers on calibration stickers at Colac WTP did not always align with the instrument number in SCADA (instrument numbers that do not match in the field and on SCADA can result in incorrect assumptions regarding the location of the instrument). OFIs (OFI 1E-1 and OFI 1E-2) have been captured to address these noncompliances.	OFI 1D-1: For completeness, add Apollo Bay WTP GAC Filtration CCP to the CCP Locations list. OFI 1D-3: Undertake a review, and revision as necessary, of all CCPs, and implement the revisions. OFI 1E-1: Develop and implement processes to improve understanding of and adherence to the chemical delivery requirements by operational staff, including the importance of record keeping. OFI 1E-2: Undertake a systems-wide review of the annual calibration and implement the findings.
2 Risk Management Plan Contents	Regulation 8(1)(i) requires the development and maintenance of appropriate critical control points, and procedures for their implementation to ensure protection of public health outcomes. Colac WTP fluoridation critical limit was inconsistent between CCP documentation (1.5 mg/L) and SCADA (1.6 mg/L). This is considered a minor noncompliance as there are more conservative shutdown processes prior to the CCP limit. While an OFI (OFI 2M-3) has been captured for this noncompliance, it is noted that the limit has already been changed in SCADA.	OFI 2M-3: The Colac fluoridation CCP limit in the SCADA should be corrected to 1.5 mg/L (this was advised as completed on 2023-03-22).
3 Identification and Management of Risks to Water Supply	None	None
4 Risk Management Plan Documents	Based on a review of all supporting documentation including at the site visits, we were largely able to confirm compliance for all requirements of s. 10(2) of the Act, with the exception of an isolated incident for chemical receipts. Chemical delivery records were not available for the entire audit period at Colac WTP (older records had been thrown out as part of a tidy-up). This fact meant that the auditors were unable to confirm the audit guidance component: "During the audit all documentation must be available for inspection." [our emphasis]. This outcome is considered minor, because it was not found to be a systemic issue. An OFI captured under Area 1 (OFI 1E-1) covers the identified noncompliance.	See OFI 1E-1.

Ms Suzie Sarkis
Manager Water
Victorian Public Health Division
Department of Health
14 / 50 Lonsdale Street
Melbourne Victoria 3000

2023-06-06

Dear Ms Sarkis

Audit Correction

I confirm that the submitted risk management plan audit certificate for Barwon Water dated 22 April 2023 contained an incorrect section reference to the *Safe Drinking Water Act 2003* (Vic).

The incorrect section reference was: **section 8(1)**

The corrected section reference should be: **section 7(1)**

Please treat the certificate as having been amended accordingly.

Yours sincerely



Dr Annette Davison
HND, BSc(Hons), MEM(LGov/Lan), PhD, PMANZ



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